

# Roberto Trotta

MSc(Hons), PhD, FRAS, SFHEA | DOB: 29.04.1977 | SISSA, Trieste, Italy  
rtrotta@sisssa.it | robertotrotta.com | ORCID | LinkedIn | iNSPIRE: R.Trotta.1

## Current and Previous Positions

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### Current

- 2024– **Director, Interdisciplinary Lab (ILAS)**  
International School for Advanced Studies (SISSA), Trieste, Italy
- 2023– **Full Professor of Theoretical Physics (Professore di I fascia)**  
Physics Department, SISSA, Trieste, Italy

### Previous Academic Positions

- 2021–25 **Head, Theoretical and Scientific Data Science group**  
SISSA, Trieste, Italy
- 2019–22 **Full Professor of Astrostatistics (on leave 2020–22)**  
Imperial College London, UK
- 2015–20 **Director, Centre for Languages, Culture and Communication**  
Imperial College London, UK
- 2008–19 **Lecturer (2008–12), Senior Lecturer (2012–16), Reader (2016–19)**  
Physics Department, Imperial College London, UK
- 2006–08 **Junior Fellow**  
St Anne's College, University of Oxford, UK
- 2005–08 **Sir Norman Lockyer Research Fellow of the Royal Astronomical Society**  
Astrophysics Department, University of Oxford, UK
- 2004–05 **Postdoctoral Researcher**  
Department of Theoretical Physics, University of Geneva, Switzerland

### Education and Qualifications

- 2022, 2020 **Abilitazione Scientifica Nazionale a professore di I fascia**  
Settori FIS 02/A2 (valid until 2032) and FIS 02/C1 (valid until 2029)
- 2011 **Certificate of advanced study in learning and teaching**  
Centre for Higher Education Research and Scholarship, Imperial College London
- 2004 **PhD in Theoretical Physics**  
University of Geneva, Switzerland. Supervisor: Ruth Durrer
- 2001 **MSc(Hons) in Physics**  
Swiss Federal Institute of Technology (ETHZ), Zurich, Switzerland

## Selected Prizes and Awards

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**Annie Maunder Medal for Outreach**, Royal Astronomical Society (2020) — the highest RAS recognition for public engagement; **President's Award for Leadership in Public Engagement**, Imperial College London (2018); **Chaire Georges Lemaître** (2018), Université Catholique de Louvain; **Royal Society Significance Lecture** (2017); **Outstanding Performance Award**, Imperial College (2017); **President's Award for Excellence in Teaching**, Imperial College London (2016); **Foreign Policy 100 Global Thinkers** (2014); **Michelson Postdoctoral Lectureship Prize**, Case Western Reserve University (2008); **Lord Kelvin Award Lecture**, British Association for the Advancement of Science (2007).

## Visiting Academic Positions

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- 2023– **Visiting Professor of Astrostatistics**  
Imperial College London, UK
- 2023– **Visiting Faculty, Data-Driven Management**  
MIB Trieste School of Management, Italy

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|---------|----------------------------------------------------------------------------------------------------------------------|
| 2019–22 | <b>Visiting Professor of Cosmology</b><br>Gresham College, London, UK                                                |
| 2011–   | <b>Next Einstein Visiting Scholar</b><br>African Institute for Mathematical Sciences (AIMS), Cape Town, South Africa |

## Teaching and PhD Supervision

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**Teaching at SISSA (2020–).** Bayesian Methods I & II; Bayesian Inference and Machine Learning in Cosmology; Ethical Aspects of AI (Theoretical and Scientific Data Science PhD); AI and Information (Master in Science Communication); Machine Learning for Business Analytics (MBA, MIB Trieste).

**Teaching at Imperial College London (2008–2020).** Lecturer of ‘Physics of the Universe’ (3rd-year core) and ‘Statistics of Measurement’ (2nd-year core); Cosmology (associate lecturer); 30+ international postgraduate schools and masterclasses (Saas-Fee, ICTP, Cargèse, Heidelberg, Niels Bohr Institute, AIMS, ESAC).

**PhD Supervision.** 9 completed PhD theses (incl. Springer Thesis Prize, two Winton Capital Best PhD Thesis prizes, RSS Emerging Applications Thesis Prize 2025); 2 ongoing PhDs; 14 postdocs (alumni now faculty at Cambridge, Uppsala, CONICET, Lawrence Livermore); 13 MSc/MSci students; 35 BSc projects; 18 external PhD/MSc examinations internationally.

## National and International Collaborations and Networks

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**Experimental Collaborations.** LSST Dark Energy Science Collaboration, member (2025–); XLZD Consortium for direct dark matter detection, member (2022–); DARWIN dark matter experiment, Executive Board member (2015–) and Science Working Package leader (2014–); ATLAS Collaboration, Short Term Associate (2014–16); XENON100, Associated Scientist (2014–18); Fermi-LAT, Affiliated Scientist (2012–); Euclid ESA mission, Theory Working Group member (2010–), likelihood Work Package co-lead (2020–22).

**Networks and Consortia.** Centro Nazionale FAIR (PNRR), Spoke Edge and Exascale Computing (2022–); Centro Nazionale HPC (PNRR), Spoke COSMOS (2022–); Programme Committee, ML and AI for Science, FVG Data Science Institute (2022–).

**Advisory and Coordination.** Scientific Consultant, ICTP Trieste (2022–); Co-lead, Astrostatistics section of EuCAPT White Paper (2021); Astrostatistics section lead, Euclid Theory Working Group (2013, 2018); Co-author, SuperBayeS public code (2006–15); Scientific Advisor on Dark Energy probes, STFC UK (2005–07).

**International Partnerships.** Active collaborations across the UK (Cambridge, Imperial, Edinburgh), USA (KITP/UCSB, Penn State, Lawrence Livermore), Switzerland (Geneva, Bern, ETHZ), the Netherlands (Amsterdam), Spain (IFT Madrid, Valencia), South Africa (AIMS, Stellenbosch) and India (IUCAA Pune).

## Other Work Experience and Consultancy

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**Advisory Roles.** Board of Directors, Fondazione Internazionale Trieste per il Progresso e la Libertà delle Scienze (2026–); Scientific Advisory Board, FACTA.eu (2025–); Scientific Advisory Board, RACHAEL S.r.l. (2024–; previously Chairman of SAB 2021–24); Consultant, Imperial Consultants (2010–).

**Entrepreneurship.** Co-founder and Director, *Data Fusion Consultants Ltd* (Imperial College spin-off, 2012–20); Non-executive Director, Imperial Consultants (2018–20).

**Industry and Media Clients (Selection).** Beluga Group, Savio Macchine Tessili, Oxford University Press, Nature Publishing Group, Red Planet Pictures (BBC ‘Death in Paradise’), Left Bank Pictures (Netflix ‘Origin’), Fantastic Productions (Netflix), BBC Sky at Night, London Science Museum, Deutsches Museum Munich, Museo del Castello di Miramare, Regione Friuli Venezia Giulia, Institute of Physics. Statistical consultancy for clients in shipping, energy, property, banking, fashion and online commerce.

**Executive Training.** Imperial College Business School Executive Education (Sberbank Digital Technology, 2020–21; Naspers Executives Programme, 2019); Generali Business Translators (2022); CINECA Academy (2022); Intesa San Paolo (2024); FINECO Advisory Seminar keynote (2025); ML for Business Analytics, MBA, MIB Trieste (2023–).

## Administrative Roles and Responsibilities

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**SISSA, Trieste.** Director, Interdisciplinary Lab (ILAS, 2025–); Head of Data Science (2021–25); PhD Coordinator and Academic Board Chair, Theoretical and Scientific Data Science (2021–25); Head of Group, Theoretical and Scientific Data Science (2021–25); Member, Physics Area Steering Board (2021–25); Knowledge Transfer Committee representative (2021–); Chair, PhD admissions panel (2021–25); Physics Area coordinator for the 2014–19 national assessment exercise (VQR, 2021); Deputy SISSA representative on Italian Computing and Data Infrastructure (2020–).

**Imperial College London.** Director, Centre for Languages, Culture and Communication (CLCC, 2015–20); member of Imperial College Senate (2016–20); Education Strategy and Operations Group (2020); Education Office COVID-19 Response Group (2020); Horizons Stream Lead, I-Explore programme (2019–20); Chair, Curriculum Review Reference Panel (2018–19); Chair of CLCC Departmental Teaching Committee, CLCC Management Team, Imperial Horizons Team, and President’s Award review panel (all 2015–20); co-chair, CLCC/CHERS Education Committee (2016–20).

**Gresham College, London.** Member of Academic Board (2019–22).

**International.** Member, EuCAIF Steering Board, European Coalition for AI in Fundamental Physics (2024–).

**Leadership Training.** Imperial Leadership and Management Development Programme (validated by the Chartered Management Institute, UK), 2015–21.

## Scientific Organisations and Coordination of Academic Activities

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**International Executive and Steering Boards.** International Executive Committee, INSAP conference series (2017–); Council of the International Astrostatistics Association (2015–); Outreach and Knowledge Transfer Leader, IAA (2015–); Astrostatistics Committee, International Statistical Institute (2011–); Working Group on Equity and Inclusion, Commission C1, IAU (2016–).

**Conference and School Organisation (Selection from 40+ Events).** LOC Chair, Convegno Nazionale di Comunicazione della Scienza, Trieste (2025); SOC, EuCAIFCon (Amsterdam 2024; Cagliari 2025); SOC, INSAP XI–XIII (CalTech 2022, Corfu 2024, Belfast 2025); SOC, Advanced Schools on Applied Machine Learning, ICTP (2024, 2026); SOC, Advanced School on Foundation Models for Scientific Discovery, ICTP (2025); Chair and founder, ‘Cosmostat’ conference series (Switzerland 2009; Banff 2013); Lead Organiser, KITP programme ‘Multi-probe Identification of Dark Matter’, UCSB (2013).

**Evaluation and Assessment Panels.** Evaluator: Agence Nationale de la Recherche, France (2026); Swiss National Science Foundation (2024, 2025); ERC Consolidator Grants (2026); ERC Synergy Grants (2025); ERC Starting Grants (2021); UKRI Future Leaders (2025); Dutch Research Council (2024); Canada Research Chairs programme (2020), South African National Research Foundation (2007–). Promotion and tenure assessor: Cambridge University (2025); U. Minnesota (2023). Panel member: STFC Rutherford Fellowships (2014–15); Royal Society International Grants (2007–12); Dutch Research Council ORC programme (2024, 2026).

## Editorial Activity

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**Editorial Boards.** Scientific Editor and founding member of the inaugural editorial board of *RAS Techniques and Instruments* (RASTI), open-access journal of the Royal Astronomical Society (2021–). Member, *Astronomy and Geophysics* Editorial Board (2009–).

**Guest Editor.** *Statistical Analysis and Data Mining* (Wiley), special issue on Astrostatistics with A. Gandy, vol. 6, 1–2 (2013); special issue, *Physics of the Dark Universe* (Elsevier, 2013).

**Peer Review.** 80+ papers refereed for *Nature Physics*, *Physical Review Letters*, *Physical Review D/E*, *ApJ*, *MNRAS*, *JCAP*, *JHEP*, *Physics Letters B*, *New Astronomy*, *Computer Physics Communications*, *Science and Education*, *RASTI*.

**Major Community and White Papers.** *Nature Astronomy* opinion piece (2025): ‘The indiscriminate adoption of AI threatens the foundations of academia’; co-lead of the Astrostatistics section, EuCAPT White Paper (2021); Astrostatistics section lead, Euclid Theory Working Group ‘*Cosmology and Fundamental Physics with the Euclid Satellite*’, Living Reviews in Relativity (2013, 2018).

## Membership of Scientific Societies

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**Elected Fellow.** International Astrostatistics Association (2016); Royal Astronomical Society (FRAS); Royal

Astronomical Society Dining Club (2012).

**Academic / Senior Fellow.** Imperial College London Data Science Institute (2018–); Higher Education Academy, UK (SFHEA, 2021).

**Professional Member.** International Society for Bayesian Analysis; International Astronomical Union; Swiss Society for Astronomy and Astrophysics; European Astronomical Society; International Statistical Institute — Astrostatistics Committee (2011–); Gresham Society.

**Affiliate.** Chartered Management Institute, UK (2021–).

**Jury and Awards Panels.** Jury Member, Premio Cosmos (Italian national science book award, 2022–); International FameLab semi-final judge, Cheltenham (2015).

## Public Engagement and Science Communication

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**Books for the General Public.** *Starborn: How the Stars Made Us* (Basic Books, 2023): BBC Radio 4 Book of the Week, BBC Sky at Night 5-star review, *New Scientist* ‘22 Best Non-Fiction and Popular Science Books of 2023’, *Smithsonian Magazine* ‘Ten Best Science Books of 2023’, 2025 Nautilus Book Awards Gold Winner; translated into Italian, Korean, Spanish and Chinese. *The Edge of the Sky* (Basic Books, 2014): cosmology in the 1,000 most common English words; for which I was named *Foreign Policy* ‘Global Thinker 2014’; translated into Korean, German and Catalan.

**Distinctions.** Annie Maunders Medal for Outreach, Royal Astronomical Society (2020); Visiting Professor of Cosmology, Gresham College London (2019–22); STFC Public Engagement Fellow (2013–17).

**Public Reach.** 198+ public lectures and talks at major science festivals (Cheltenham, Edinburgh, New Scientist Live, Royal Albert Hall, Royal Institution, Mantova, Torino, Genova); 47+ articles for the general public (*TIME*, *The Guardian*, *New Scientist*, *La Stampa*, *Le Scienze*); broadcast appearances on BBC Radio 4, BBC World Service, NPR, RAI, RSI Swiss National Radio/TV, Sky TG24.

**Art and Science Projects.** Co-author and creative director, *LIBRA* (2021, multimedia theatre play, performed at Miramare Castle, Reggio Calabria, Gorizia 2025 European Capital of Culture); AI consultant, *Parl-IA-moci* (2023, theatre play with Diana Höbel and ChatGPT); *A Multi-Sensorial Dark Matter Experience* (2019); *Dinner with Copernicus* (2018); *g-ASTRONOMY* multi-sensorial cosmology dinners; *The Theory of Everything* short film (audience award, International Short Film Festival Berlin 2013).

**Consulting for Film and Media.** Scientific consultant for Red Planet Pictures (BBC ‘Death in Paradise’), Left Bank Pictures (‘Origin’, Netflix), Fantastic Productions (Netflix feature film), Nature Publishing Group, Oxford University Press.

## Bibliographic Indicators

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**Output.** 115 papers in peer-reviewed physics journals or machine learning conferences (12 as first author, 3 in *Phys. Rev. Lett.*, 1 in *Nat. Astron.*); 18 conference proceedings; 5 white papers and editorial works; 5 book chapters. The single-author review ‘*Bayes in the sky*’ (Contemporary Physics, 2008) has 1500+ citations.

**Citations (July 2026).** 14,000+ citations, h-index 53 ([INSPIRE](#)); 15,000+ citations, h-index 59 ([Google Scholar](#)). Citation thresholds: 16 papers >100 citations; 4 papers >250; 2 papers >500; 3 papers >1000. *Full publications list given in Part II.*

## Part II — Detailed Record

*Full lists of appointments, details on grants, publications, knowledge transfer activities and public engagement.*

### ACADEMIA

#### Academic Appointments (full list)

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|---------|---------------------------------------------------------------------------------------------------------------|
| 2024–   | Director, Interdisciplinary Lab (ILAS), SISSA, Trieste                                                        |
| 2023–   | <a href="#">Professor of Theoretical Physics</a> (Professore di I fascia), Physics Department, SISSA, Trieste |
| 2021–25 | Head of Data Science, SISSA, Trieste                                                                          |
| 2020–22 | Associate Professor (Professore di II fascia), Physics Department, SISSA, Trieste                             |
| 2019–22 | <a href="#">Professor of Astrostatistics</a> , Imperial College London                                        |
| 2015–20 | Director, <a href="#">Centre for Languages, Culture and Communication</a> , Imperial College London           |
| 2016–19 | Reader in Astrophysics (equivalent to Associate Professor), Imperial College London                           |
| 2013–17 | Public Engagement Fellowship, Science and Technology Facilities Council, UK                                   |
| 2012–16 | Senior Lecturer in Astrophysics, Imperial College London                                                      |
| 2008–12 | Lecturer in Astrophysics (equivalent to Assistant Professor), Imperial College London                         |
| 2006–08 | Junior Fellow, St Anne’s College, Oxford                                                                      |
| 2006    | Junior Fellow Linacre College, Oxford (declined)                                                              |
| 2005–08 | Sir Norman Lockyer Research Fellow of the Royal Astronomical Society, University of Oxford                    |
| 2005    | European Space Agency Research Fellowship (declined)                                                          |
| 2004–05 | Research Associate, University of Geneva, Switzerland                                                         |

#### Honours and Awards

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| 2022 | Charmian Brinson Honorary Lecture, Imperial College London                                                                                                                                                                             |
| 2020 | Annie Maunder Medal for Outreach, Royal Astronomical Society                                                                                                                                                                           |
| 2018 | Chaire Georges Lemaître, Université Catholique de Louvain, Belgium; President’s Award for Leadership in Public Engagement, Imperial College London                                                                                     |
| 2017 | Significance Lecture, Royal Statistical Society; Outstanding Performance Award, Imperial College London; Highly commended, President’s Award for Excellence in Societal Engagement, Imperial College London                            |
| 2016 | Longlisted, NCCPE Engage Competition Award (individual project category); President’s Award for Excellence in Teaching, Imperial College London; Faculty of Natural Sciences Prize for Excellence in Teaching, Imperial College London |
| 2015 | Excellence Award, Stio town hall council, Salerno, Italy                                                                                                                                                                               |
| 2014 | Foreign Policy 100 Global Thinkers                                                                                                                                                                                                     |
| 2012 | Elected to the Royal Astronomical Society Dining Club                                                                                                                                                                                  |
| 2011 | Merit Award, Imperial College London                                                                                                                                                                                                   |
| 2008 | Michelson Postdoctoral Lectureship Prize, Case Western Reserve University, USA                                                                                                                                                         |
| 2007 | Lord Kelvin Award Lecture, British Association for the Advancement of Science; Merit Award, Royal Astronomical Society; Merit Award, Oxford University                                                                                 |

#### Distinguished Fellowships and Professional Memberships

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**Elected Fellow.** International Astrostatistics Association (2016); Royal Astronomical Society (FRAS); Royal Astronomical Society Dining Club (2012).

**Academic / Senior Fellow.** Imperial College London Data Science Institute (2018–); Higher Education Academy, UK (Senior Fellow, 2021; Fellow status achieved 2011).

**Professional Member.** International Society for Bayesian Analysis; International Astronomical Union; Swiss Society for Astronomy and Astrophysics; European Astronomical Society; Gresham Society.

**Affiliate.** Chartered Management Institute, UK (2021–2022).

## Continuing Professional Development

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| 2015–21 | Imperial Leadership and Management Development Programme (validated by the Chartered Management Institute, UK). Courses on: introduction to management; managing groups and teams; leading successful teams; effective recruitment and selection; finance for non-finance managers; promoting equality and diversity; CPD portfolio; 360 review and action planning; decision making; unconscious bias; managing difficult situations. |
| 2016    | Introduction to IP rights                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2014    | Bond Solon Expert Witness training programme                                                                                                                                                                                                                                                                                                                                                                                           |

## Scientific Editor

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| 2021– | Scientific Editor and founding member of the inaugural editorial board of <i>RAS Techniques and Instruments</i> (RASTI), the open-access journal of the Royal Astronomical Society covering data collection methods, instrumentation, data processing and modelling, AI and machine learning, statistical methods, software, and high-performance computing. |
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## Scientific Activities

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| 2026–      | Management Committee Member, EU Cost Action ‘Artificial Intelligence for Fundamental Physics: Open Roadmapping and Inclusive Actions (AIFORIA)’                                                                                |
| 2025–      | Member, LSST Dark Energy Science Collaboration                                                                                                                                                                                 |
| 2022–      | Scientific Consultant, International Centre for Theoretical Physics (ICTP), Trieste; Programme Committee Member, ‘ML and AI for Science’, FVG Data Science Institute; Member, XLZD Consortium for direct dark matter detection |
| 2014–      | DARWIN direct dark matter detection experiment: Executive Board member (2015–), Science Working Package leader (2014–)                                                                                                         |
| 2014–16    | ATLAS Collaboration, Short Term Associate                                                                                                                                                                                      |
| 2014–18    | XENON100 direct dark matter detection experiment, Associated Scientist                                                                                                                                                         |
| 2012–      | Fermi-LAT Affiliated Scientist                                                                                                                                                                                                 |
| 2012, 2013 | Guest editor: <i>Statistical Analysis and Data Mining</i> special issue (2012); <i>Physics of the Dark Universe</i> special issue (2013)                                                                                       |
| 2010–      | Euclid ESA space mission, Theory Working Group member; likelihood analysis Work Package co-lead (2020–22)                                                                                                                      |
| 2005–07    | Scientific Advisor on Dark Energy probes, STFC, UK                                                                                                                                                                             |
| 2006–15    | Co-author of the <b>SuperBayeS</b> public code for the analysis of supersymmetric theories                                                                                                                                     |
| 2001–      | 30+ invited scientific talks, 110+ research talks and seminars                                                                                                                                                                 |

## Service

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| 2026      | Dutch Research Council evaluator, ERC Consolidator Grants evaluator                                                                                                                     |
| 2025      | Emmanuel College Cambridge Research Fellowship assessor; Cambridge University promotion assessor; UKRI Future Leaders evaluator; ERC Synergy Grants evaluator                           |
| 2024      | Dutch Research Council evaluator; Swiss National Science Foundation evaluator                                                                                                           |
| 2023      | Tenure committee reviewer, School of Statistics, University of Minnesota                                                                                                                |
| 2021      | ERC Starting Grants evaluator                                                                                                                                                           |
| 2020      | Referee, Canada Research Chairs programme                                                                                                                                               |
| 2018      | Member, STFC Public Engagement Early Career Researcher Network Review Panel                                                                                                             |
| 2016–     | Member, Working Group on Equity and Inclusion, Commission C1 of the IAU                                                                                                                 |
| 2015–     | Member, DARWIN-LXe Executive Board; Outreach and Knowledge Transfer Leader, International Astrostatistics Association; Member, Council of the International Astrostatistics Association |
| 2015      | University Research Fellowships referee, Royal Society; Imperial College London representative, Interdisciplinary Curriculum Group                                                      |
| 2014–15   | STFC Rutherford Fellowships panel                                                                                                                                                       |
| 2011–     | Member, Astrostatistics Committee, International Statistical Institute                                                                                                                  |
| 2010      | Founding member, Imperial Centre for Inference and Cosmology                                                                                                                            |
| 2009      | Jesus College and Peterhouse College fellowships referee, Cambridge                                                                                                                     |
| 2009–2012 | Member, <i>Astronomy &amp; Geophysics</i> Editorial Board                                                                                                                               |
| 2007–     | South African National Research Foundation evaluator                                                                                                                                    |
| 2007–12   | Royal Society International Grants Panel                                                                                                                                                |

Ongoing Referee for *Nature Physics*, *PRL*, *ApJ*, *JCAP*, *JHEP*, *PLB*, *Phys. Rev. D/E*, *New Astron.*, *MNRAS*, *Comp. Phys. Comm.*, *Science and Education*, *RASTI* — 80+ papers refereed.

## Institutional Responsibilities, Management and Leadership

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### SISSA, Trieste

2025– Director, Interdisciplinary Lab (ILAS)  
2021–25 PhD Coordinator and Academic Board Chair, Theoretical and Scientific Data Science; Head of Group, Theoretical and Scientific Data Science; Member, Giunta dell'Area Fisica (Physics Area Steering Board)  
2021– Member, Laboratorio Interdisciplinare per le Scienze Naturali (ILAS); Physics Area representative, Knowledge Transfer Committee (Commissione per la Valorizzazione del Talento e il Trasferimento della Conoscenza)  
2021 Physics Area coordinator for the 2014–19 national assessment exercise (VQR)  
2020–25 Deputy SISSA representative, Italian Computing and Data Infrastructure general assembly

### Gresham College, London

2019–22 Member of the Academic Board

### International

2024– Member, EuCAIF Steering Board, European Coalition for AI in Fundamental Physics

### Imperial College London

2020 Education Strategy and Operations Group  
2019–20 Horizons Stream Lead, I-Explore programme  
2020 Education Office COVID-19 Response Group  
2018–19 Chair, Curriculum Review Reference Panel  
2015–20 Chair, CLCC Departmental Teaching Committee  
2016–20 Chair, President's Award review panel 'Supporting the Student Experience'  
2015–20 Chair, Imperial Horizons Team; Chair, CLCC Management Team  
2016–20 Co-chair, CLCC/CHERS Education Committee  
2018–20 Member, I-Explore Modules Innovation Group; Member, Enterprise Academic Advisory Group; Member, Engagement Strategy Group  
2015–20 Member, Programmes Committee  
2013–20 Member, Imperial SpaceLab Steering Group  
2016–20 Member, Strategy Board, Centre for Performance Science  
2017–20 Member, Online Learning Innovation Group  
2016–18 Member, Academic Partnerships Engagement with Research Steering Group  
2015–20 Member, Education Office Management Committee  
2016–20 Member, Imperial College Senate  
2015–18 Member, Imperial Horizons Committee  
2015–17 Member, Task and Finish Group for Assessment

## Organising of Scientific Meetings

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2026 SOC, 'DESC Virtual Collaboration Meeting' (online)  
▷ LOC, 'Quasar Bazar', IFPU, Trieste  
▷ SOC, Advanced School in Applied Machine Learning: Uncertainty and Robustness of DL Models for Science, ICTP  
▷ SOC and co-proposer, NeurIPS workshop 'AI and Science: Evolution or Extinction?'

2025 LOC Chair, Convegno Nazionale di Comunicazione della Scienza, Trieste  
▷ SOC, Advanced School on Foundation Models for Scientific Discovery, ICTP, Trieste  
▷ SOC, AI & Physics track, 'AI4Science' conference, Ljubljana  
▷ SOC, AI Alliance Meet-up, Trieste  
▷ SOC, INSAP XIII conference, Belfast  
▷ SOC, EuCAIFCon 2025, Cagliari

2024 SOC, Advanced School on Applied Machine Learning, ICTP, Trieste  
▷ SOC, StatPhys conference, Imperial College London  
▷ SOC, EuCAIFCon 2024, Amsterdam  
▷ SOC, INSAP XII conference, Corfu

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|            | <ul style="list-style-type: none"> <li>▷ LOC, ‘Unsupervised Exploration of Fast Radio Bursts Dynamic Spectra’, IFPU Team Research Workshop, Trieste</li> <li>▷ SOC, ‘Statistics meets ML’, Imperial College London</li> <li>▷ SOC, COSMO21 conference, Chania</li> </ul>                                                                                                                                     |
| 2023       | Organiser, ‘AI Interactions: ChatGPT and Beyond’, Trieste and online <ul style="list-style-type: none"> <li>▷ SOC, ‘Machine Learning Enabled Searches for New Physics in Astrophysical Data’, IFPU, Trieste</li> </ul>                                                                                                                                                                                       |
| 2022       | Convenor and moderator, Year of Basic Sciences for Sustainable Development: Embedding Ethics in Machine Learning, ICTP, Trieste (online) <ul style="list-style-type: none"> <li>▷ SOC, ‘Ethical and Societal Challenges of Machine Learning’, ICTP, Trieste (online)</li> <li>▷ SOC, ‘Mind the Gap: How We Build, Perceive, Experience and Share Spaces’, SISSA</li> <li>▷ SOC, INSAP XI, CalTech</li> </ul> |
| 2017–      | International Executive Committee, ‘Inspiration of Astronomical Phenomena’ (INSAP)                                                                                                                                                                                                                                                                                                                           |
| 2018       | SOC, Astronomical Data Analysis 9, Valencia                                                                                                                                                                                                                                                                                                                                                                  |
| 2016       | SOC, Astronomical Data Analysis 8, Crete                                                                                                                                                                                                                                                                                                                                                                     |
| 2015       | Co-Chair, ‘Identification of Dark Matter with a Cross-Disciplinary Approach’ workshop, UAM Madrid                                                                                                                                                                                                                                                                                                            |
| 2014       | Chair, ‘From Cosmology to Customers: Astrostatistics Solutions to Big Data Problems’ industry-academia workshop, Imperial                                                                                                                                                                                                                                                                                    |
| 2009, 2013 | Chair and founder, ‘Cosmostat’ conference series: Centro Stefano Franscini, Switzerland (2009) <ul style="list-style-type: none"> <li>▷ Banff International Research Station (2013)</li> </ul>                                                                                                                                                                                                               |
| 2013       | Lead Organiser, ‘Multi-probe Identification of Dark Matter’, Kavli ITP programme, UCSB                                                                                                                                                                                                                                                                                                                       |
| 2012       | Chair, DarkAttack12, Centro Stefano Franscini, Switzerland <ul style="list-style-type: none"> <li>▷ SOC, ICIC inaugural workshop, Imperial College London</li> <li>▷ SOC, ‘Statistical Issues in Searches’, SLAC, Palo Alto</li> </ul>                                                                                                                                                                       |
| 2010       | SOC, RAS National Astronomy Meeting, Glasgow <ul style="list-style-type: none"> <li>▷ Co-chair, RAS Specialist Discussion Meeting ‘Novel Methods for the Exploitation of Large Astronomical and Cosmological Data Sets’, London</li> <li>▷ Co-chair, IOP dark matter meeting, London</li> <li>▷ Co-chair, Astrostatistics workshop, London</li> </ul>                                                        |
| 2012, 2013 | Founder and chair, ‘The Big Questions’ public lecture series (2012) and ‘The Sensual Universe’ public lecture series (2013), London                                                                                                                                                                                                                                                                          |
| 2008       | Co-chair, ‘Bayesian Evidence’ workshop, Brighton                                                                                                                                                                                                                                                                                                                                                             |
| 2007       | Co-chair, RAS Specialist Discussion Meeting ‘Statistical Challenges in Particle Astrophysics and Cosmology’, London                                                                                                                                                                                                                                                                                          |
| 2002–05    | Co-chair, Journées de Cosmologie des Lacs Alpains, Geneva/Annecy/Lausanne/CERN                                                                                                                                                                                                                                                                                                                               |

## Teaching

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### Undergraduate Courses and Lectures

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| 2021       | Guest lunchtime lecture, ‘The Machine Learning Revolution in Cosmology’, Imperial (online) |
| 2015–20    | Lecturer, ‘Physics of the Universe’, 3rd-year physics core course, Imperial                |
| 2008–12    | Lecturer, ‘Statistics of Measurement’, 2nd-year physics core course, Imperial              |
| 2010–20    | Associate lecturer: ‘Cosmology’, Imperial                                                  |
| 2008–09    | Associate lecturer: ‘Sun, Stars and Planets’ (2008–09)                                     |
| 2017, 2018 | Guest lecturer, ‘Communicating Science’, Imperial Horizons course                          |
| 2016       | Guest lecturer, ‘Creative Writing’, Imperial Horizons course                               |
| 2010       | Guest lecturer, ‘Astrophysics’, 4th-year option course, Imperial                           |

### Postgraduate Courses at SISSA

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|---------|-------------------------------------------------------------------------------------------------------------|
| 2023–   | Bayesian Methods I, Theoretical and Scientific Data Science PhD programme                                   |
| 2020–   | Ethical Aspects of AI, Theoretical and Scientific Data Science PhD programme                                |
| 2021–   | Intelligenza Artificiale e Informazione (with Elisabetta Tola), Master in Science Communication             |
| 2020–23 | Bayesian Methods II, Theoretical and Scientific Data Science PhD programme                                  |
| 2021–   | Bayesian Inference and Machine Learning in Cosmology, Theoretical and Scientific Data Science PhD programme |

### Postgraduate Lectures, Masterclasses and Graduate Schools

|       |                                                                                     |
|-------|-------------------------------------------------------------------------------------|
| 2026  | AI training course, SISSA; EFC network AI training workshop, Bertinoro              |
| 2024– | Machine Learning for Business Analytics, MBA in International Business, MIB Trieste |

|      |                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023 | Introduction to Bayesian Inference, SISSA Master in HPC, Trieste; Machine Learning for Business Analytics, MBA in International Business, MIB Trieste                                                                                                                                                                                                                                                |
| 2022 | Introduction to Data Science, International Master in Physics of Complex Systems, ICTP, Trieste; Statistics lectures, Cosmology Summer School, ICTP, Trieste                                                                                                                                                                                                                                         |
| 2021 | Avventure fra Arte e Scienza nel Comunicare la Cosmologia, SISSA MSC lecture; From Bayes to Machine Learning, Heidelberg Grad Days (online); Bayesian Inference, Saas-Fee Course on ‘Astronomy in the Era of Big Data’ (online)                                                                                                                                                                      |
| 2019 | Keynote, Research Computing Summer School, Imperial College London                                                                                                                                                                                                                                                                                                                                   |
| 2018 | ICIC Data Analysis Postgraduate School, Imperial College London; Analytics, Inference and Computation in Cosmology: Advanced Methods, Cargèse, France; Astronomical Data Analysis IX Summer School, Valencia, Spain                                                                                                                                                                                  |
| 2017 | Stats4Astro Astrostatistics School, Autrans, France; ISAPP Summer School, Texel, The Netherlands; First Italian Astrostatistics School, Milan                                                                                                                                                                                                                                                        |
| 2016 | European Space Astronomy Centre (ESAC) Statistics Workshop, Villanueva de la Cañada, Spain; ADA8 Summer School on Astronomical Data Analysis, Chania, Greece                                                                                                                                                                                                                                         |
| 2014 | XII School of Cosmology, Cargèse, France; Niels Bohr Institute PhD School, ‘Statistical Methods’, Copenhagen; XXVI Winter School, ‘Bayesian Inference in Astronomy and Astrophysics’, Canary Islands; ICIC STFC Data Analysis School, London; 44th Saas-Fee Advanced Course on Astronomy and Astrophysics, ‘Cosmology with Wide-Field Surveys’, Switzerland                                          |
| 2013 | School on Bayesian Analysis in Physics and Astronomy, Stellenbosch, South Africa; ICIC Data Analysis Postgraduate School, London                                                                                                                                                                                                                                                                     |
| 2012 | ‘Statistical Inference’, African Institute for Mathematical Sciences postgraduate course, Cape Town                                                                                                                                                                                                                                                                                                  |
| 2011 | ‘Advanced Statistical Methods for Cosmology and Astroparticle Physics’, Niels Bohr Institute, Copenhagen                                                                                                                                                                                                                                                                                             |
| 2010 | Cosmology postgraduate lectures, Imperial; ‘Advanced Statistical Methods’ lectures, Imperial; ‘Statistical Methods for Cosmology’, 1st Jayme Tiomno School of Cosmology, Rio de Janeiro                                                                                                                                                                                                              |
| 2009 | ‘Advanced Statistical Tools for Cosmology’, Valencia University; ‘Probability Theory, Statistics and Data Analysis’ postgraduate lectures, Imperial/UCL; ‘The CMB and its Statistical Interpretation’, Würzburg University postgraduate lecture series; ‘Modern Cosmology’ postgraduate lectures, Imperial/UCL; ‘Advanced Statistical Tools for Cosmology’ postgraduate lectures, U. Autónoma Madrid |

## Supervision

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### Postdoctoral Researchers

|         |                                                                               |
|---------|-------------------------------------------------------------------------------|
| 2025–   | Giulio Scelfo                                                                 |
| 2024–   | Junsong Cang                                                                  |
| 2024–25 | Martin de los Rios (now faculty at CONICET, Argentina); David Prelogović      |
| 2023–   | Rahul Srinivasan                                                              |
| 2023–25 | Chiara Moretti                                                                |
| 2023    | Matteo Breschi                                                                |
| 2022–   | Andre Scaffidi                                                                |
| 2022–24 | Gabriella Contardo                                                            |
| 2020–22 | Ira Wolfson                                                                   |
| 2020–21 | Victor Bonjean                                                                |
| 2018–20 | Eliel Camargo-Molina (now faculty at U. Uppsala)                              |
| 2017–20 | Alex Geringer-Sameth (now permanent staff at Lawrence Livermore National Lab) |
| 2014    | Kaisey Mandel (now faculty at Cambridge)                                      |

### Current PhD and MSc Students

|       |                                                   |
|-------|---------------------------------------------------|
| 2023– | M. Rigo, S. Mishra, M. Giunta, U. Zivanovic (MSc) |
|-------|---------------------------------------------------|

### Completed PhD Students (Thesis Title; Distinctions and First Destination)

|         |                                                                                                                                                                                                                                                                                                                                    |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021–25 | M. Geng (SISSA), <i>Reflections on Distributions: Humans, Machines, Languages</i> (postdoc at École Normale Supérieure, Paris)                                                                                                                                                                                                     |
| 2020–24 | K. Karchev (SISSA), <i>SNIa Cosmology for the 21st Century</i> – IAU Best Thesis Prize: Facilities, Technologies and Data Science 2026) (Simons Foundation Fellow, Barcelona)                                                                                                                                                      |
| 2020–23 | M. Autenrieth (Imperial Statistics, co-supervision), <i>Principled Bayesian Modelling and Statistical Learning with Non-Representative Data in Astrophysics</i> — Finalist, ASA Astrostatistics Interest Group 2022 Student Paper Competition; Winner, same 2024; RSS Emerging Applications Thesis Prize 2025; postdoc at Imperial |

|         |                                                                                                                                                                                                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2017–22 | W. Rahman (Imperial), <i>Constraining the Anisotropic Expansion of the Universe with Type Ia Supernovae and Improving the Treatment of Selection Effects within Bayesian Hierarchical Models</i> (data scientist at Meta) |
| 2015–19 | S. Hoof (Imperial, co-supervision), <i>Global Fits of Axions and WIMPs in Astrophysics, Cosmology and Particle Physics</i> (Imperial President Scholarship holder; postdoc at Göttingen)                                  |
| 2014–18 | J. McKay (Imperial, co-supervision), <i>Constraining Dark Matter with Renormalisation and Global Fits</i> (Imperial President Scholarship holder; software developer at Verizon Connect)                                  |
| 2013–17 | H. Shariff (Imperial), <i>Application of Bayesian Statistics in Supernovae Ia Cosmology</i> (investment consultant at Mercer)                                                                                             |
| 2011–14 | C. Strece (Imperial), <i>Characterisation of Particle Dark Matter via Multiple Probes</i> — Winton Capital best PhD thesis in physics prize; quant at CityBank                                                            |
| 2008–11 | M. March (Imperial), <i>Advanced Statistical Methods for Astrophysical Probes of Cosmology</i> — Springer Thesis Prize, Winton Capital best PhD thesis in physics prize; postdoc at Penn State                            |

### MSc / MSci Students

|      |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | U. Zivanović (U. Trieste), <i>An Application of the Rotary Masked Autoencoder to Irregular Sparse Astrophysics Datasets</i>                                                                                                                                                                                                                                                                                      |
| 2023 | R. Serra (U. Trieste), <i>Improving Photometric Galaxy Redshift Estimation under Covariate Shift with StratLearn</i>                                                                                                                                                                                                                                                                                             |
| 2022 | S. di Gioia (U. Trieste), <i>Fast and Parallel Algorithms for Causal Discovery in Astrophysics</i>                                                                                                                                                                                                                                                                                                               |
| 2021 | R. Corti (U. Trieste), <i>Machine Learning Methods for Supernova Type Ia Selection Effects Modelling</i>                                                                                                                                                                                                                                                                                                         |
| 2019 | M. von Wietersheim-Kramsta (Imperial), <i>Measuring the Galaxy Mass Function Using Photometric Data</i>                                                                                                                                                                                                                                                                                                          |
| 2017 | H. Bouvier (Imperial), <i>Determining Type Ia Supernovae Host Galaxy Characteristics Using Galaxy Zoo: Hubble and the PROSPECTOR-<math>\alpha</math> Model</i>                                                                                                                                                                                                                                                   |
| 2017 | S. Kobayashi (Imperial), <i>Determining Type Ia Supernovae Host Galaxy Characteristics Using Galaxy Zoo: Hubble and the PROSPECTOR-<math>\alpha</math> Model</i>                                                                                                                                                                                                                                                 |
| 2016 | E. Revsbech (Imperial Statistics, co-supervision), <i>Synthetically Augmented Light Curve Classification</i> — Winton Capital Prize Best Project; I. Siska (Imperial), <i>Constraining Anisotropic Cosmological Expansion using Supernovae Type Ia within a Bayesian Statistical Framework</i> ; A. Monge Imedio (Imperial), <i>Constraining Cosmological Anisotropies with SNIa Data Using Bayesian Methods</i> |
| 2015 | S. Pak (Imperial), <i>Constraining Cosmic Expansion Anisotropy using Type Ia Supernovae</i> ; K. Blanchette (Imperial), <i>Cosmological Parameter Inference in a Bayesian Hierarchical Model with Redshift-Dependent Hubble Residuals</i>                                                                                                                                                                        |
| 2011 | G. Pisani (co-supervision, Rome La Sapienza)                                                                                                                                                                                                                                                                                                                                                                     |

### Other Supervision

|         |                                                                                |
|---------|--------------------------------------------------------------------------------|
| 2011–20 | 35 BSc projects, Imperial                                                      |
|         | 22 Summer interns / UROP / IROP / ICTP / UWC undergraduate students supervised |

## Examining at Postgraduate Level

### External Examiner / Assessor

|      |                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | A. Bairagi (PhD, Institut d'Astrophysique de Paris)                                                                                                                            |
| 2025 | V. Blasone (PhD, U. Trieste, Applied Data Science and AI programme)                                                                                                            |
| 2024 | Z. Rokavec (MSc, U. Nova Gorica); A. Prajapati (PhD, Gran Sasso Science Institute); D. Prelogović (PhD, Scuola Normale Superiore; committee member); S. Rinaldi (PhD, U. Pisa) |
| 2023 | D. Lanzieri (PhD, CEA Paris-Saclay)                                                                                                                                            |
| 2022 | J. Thorp (PhD, Cambridge); F. Agocs (PhD, Cambridge)                                                                                                                           |
| 2019 | R. Hardwick (PhD, U. Portsmouth)                                                                                                                                               |
| 2018 | J. Faulkner (PhD, U. Edinburgh)                                                                                                                                                |
| 2017 | S. Hee (PhD, U. Cambridge); H. Mootoovaloo (MSc, U. Cape Town)                                                                                                                 |
| 2016 | T. Walwyn (MSc, U. Cape Town)                                                                                                                                                  |
| 2014 | F. Vansyngel (PhD, Institut d'Astrophysique de Paris)                                                                                                                          |
| 2013 | H. Silverwood (MSc, U. Canterbury, NZ)                                                                                                                                         |
| 2011 | Y. Akrami (PhD, Stockholm University; opponent); J. Newling (MSc, U. Cape Town)                                                                                                |

### Internal Examiner, Imperial College London

|      |                                                                      |
|------|----------------------------------------------------------------------|
| 2022 | E. Hauke (PhD, Centre for Higher Education Research and Scholarship) |
| 2017 | Shijing Si (PhD, Statistics Section)                                 |
| 2016 | J. Alsing (PhD, Astrophysics group)                                  |
| 2013 | H. Shariff (MSc, Astrophysics group)                                 |

## Invited Scientific Talks

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|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | Plenary talk, EUCAIFCon, Heidelberg <ul style="list-style-type: none"> <li>▷ Physics dept, Ljubljana U.</li> <li>▷ Data science seminar, CERN</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2025 | Panelist, ‘Agentic AI, LLMs and Research’, SISSA, Trieste <ul style="list-style-type: none"> <li>▷ Keynote, IFPU Focus Week, Trieste</li> <li>▷ Keynote, Randstad Foundation think tank meeting, Milan</li> <li>▷ Keynote, Ital-IA, Italian National meeting on AI, Trieste</li> </ul>                                                                                                                                                                                                                                                                                                                      |
| 2024 | Panel, Climate Conference Arena, Bologna <ul style="list-style-type: none"> <li>▷ Colloquium, Celebrating 60 Years of the Associates Programme, ICTP, Trieste</li> <li>▷ AI panel moderator, ICTP Symposium ‘The Future of Scientific Computing’, Trieste</li> <li>▷ <a href="#">Scientific Afternoons of the University of Nova Gorica</a>, Nova Gorica</li> <li>▷ CMSP News and Views Colloquium, ICTP, Trieste</li> <li>▷ Erwin Schrödinger Colloquium, University of Zürich</li> <li>▷ Keynote, Research in Dialogue, University of Bern</li> <li>▷ Heidelberg Joint Astronomical Colloquium</li> </ul> |
| 2023 | Astrophysics Colloquium, Ludwig-Maximilians-Universität, Munich <ul style="list-style-type: none"> <li>▷ Colloquium Series in Theoretical and Computational Physics (CSTCP)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2022 | ‘StratLearn: A General-Purpose Statistical Method for Improved Learning under Covariate Shift’, IAA/IAU Astroinformatics and Astrostatistics Seminars (online)                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2021 | ‘The Impact of Machine Learning in Cosmology’ (online)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2019 | Plenary, Picturing the Invisible Conference, UAL, London <ul style="list-style-type: none"> <li>▷ Keynote, ‘Artificial Intelligence: Art or Science?’, SISSA, Trieste</li> <li>▷ Keynote, ICTP, Trieste</li> <li>▷ AHRC Network meeting</li> </ul>                                                                                                                                                                                                                                                                                                                                                          |
| 2017 | Colloquium, University of Bern <ul style="list-style-type: none"> <li>▷ Colloquium, Indian Institute of Technology Madras</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2016 | Keynote, Global Challenges Showcase event, Imperial College London <ul style="list-style-type: none"> <li>▷ Keynote, COSMO21 Conference, Chania, Greece</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2015 | Southampton Physics Colloquium <ul style="list-style-type: none"> <li>▷ BASP Frontiers 2015, Villars, Switzerland</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2013 | Director’s Blackboard Talk, Kavli Institute, Santa Barbara                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2012 | Invited plenary, Statistical Issues in Searches, SLAC, Stanford <ul style="list-style-type: none"> <li>▷ Invited plenary, Astronomical Data Analysis 7, Cargèse</li> <li>▷ Invited plenary, MaxEnt 2012, Garching</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                |
| 2011 | Invited plenary, 58th ISI World Congress, Dublin <ul style="list-style-type: none"> <li>▷ Invited plenary, Statistical Challenges in Modern Astronomy V, Penn State</li> <li>▷ Invited talk, IOP dark matter meeting, London</li> <li>▷ Invited plenary, Aspen Winter Conference, ‘Indirect and Direct Detection of Dark Matter’, Aspen</li> <li>▷ Colloquium, AIMS, Cape Town</li> </ul>                                                                                                                                                                                                                   |
| 2010 | Colloquium, EPFL <ul style="list-style-type: none"> <li>▷ Invited plenary, PROSPECT workshop, Stockholm</li> <li>▷ Summary talk, ‘Statistical Issues Relevant to Significance of Discovery Claims’, Banff</li> <li>▷ Invited plenary, TeV Particle Astrophysics, Paris</li> </ul>                                                                                                                                                                                                                                                                                                                           |
| 2009 | Review talk, ILIAS/ENTApP meeting, CERN, Geneva <ul style="list-style-type: none"> <li>▷ Invited talk, National Astronomy Meeting, Hatfield</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2008 | Michelson Prize Lecture, Case Western Reserve University, USA <ul style="list-style-type: none"> <li>▷ Invited plenary, ESF Exploratory Workshop, Oporto</li> <li>▷ Invited talk, Tools08, Munich</li> <li>▷ Review talk, XXVI Workshop on Recent Developments in High Energy Physics and Cosmology, Ancient Olympia</li> </ul>                                                                                                                                                                                                                                                                             |
| 2007 | Lord Kelvin Award Lecture <ul style="list-style-type: none"> <li>▷ Invited talk, TeV Particle Astrophysics 2007, Venice</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2006 | Invited talk, 11th Marcel Grossmann Meeting, Berlin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

- 2005
  - ▷ Invited plenary, RAS General Meeting, London
- 2005 Review talk, ‘Recent Advances in Particle Physics and Cosmology’, Thessaloniki

## Contributed Conference Talks

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- 2024 The Dark Matters, parallel session at the RGS-IBG Annual International Conference (online)
- 2021 Debating the potential of machine learning in astronomical surveys (online)
- 2019 CosmoAI, Ascona, Switzerland
- 2018 Cosmo21, Valencia
- 2017 The Road to the Stars, Santiago de Compostela
- 2016 XENON100 Collaboration Meeting, Mainz, Germany
- 2013 ISI World Statistics Meeting, Hong Kong
- 2012 ISBA World Meeting, Kyoto
- 2011 ERC Kickoff meeting, Amsterdam
  - ▷ IOP Dark Matter Meeting at King’s College
  - ▷ 58th ISI World Congress
  - ▷ PHYSTAT2011, CERN, Geneva
- 2009 Corfu Summer Institute, Greece
  - ▷ cosmostats09, Ascona, Switzerland
  - ▷ Galileo Galilei Institute dark energy conference, Florence
- 2008 Nested Sampling workshop, Cambridge
  - ▷ Bayesian evidence workshop, University of Sussex
  - ▷ CosmoTools08, Marseille
- 2007 Initial Conditions in Cosmology workshop, Würzburg, Germany
  - ▷ CERN dark matter workshop, Geneva
- 2006 RAS Young Astronomers Meeting, London
  - ▷ Identification of Dark Matter Conference, Rhodes, Greece
  - ▷ Galileo Galilei Institute Cosmology Workshop, Florence
  - ▷ The Dark Side of the Universe Workshop, Madrid
  - ▷ CosmoUK Meeting, Oxford
  - ▷ Francesco Melchiorri Memorial Conference, Rome
  - ▷ TeV Particle Astrophysics II, Madison, Wisconsin
  - ▷ Astroparticle UK Meeting, Sheffield
  - ▷ National Astronomy Meeting, Leicester
  - ▷ SKA conference, Oxford
  - ▷ St Anne’s College, Oxford
- 2005 European Dark Energy Network Meeting, Paris
  - ▷ Subaru/Gemini dark energy meeting, Waikoloa, Hawaii
  - ▷ PHYSTAT05, Oxford
  - ▷ Sils Maria Cosmology Workshop, Engadin
  - ▷ ENTApP meeting, CERN, Geneva
- 2004 7th Journée des Lacs Alpins de Cosmologie, EPFL Lausanne
  - ▷ 6th Journée de Cosmologie des Lacs Alpins, Geneva
- 2003 Second Aegean Summer School on the Early Universe, Syros
  - ▷ Sils Maria Cosmology Workshop, Engadin
  - ▷ ATLAS Collaboration Physics Tutorial, CERN
  - ▷ X Marcel Grossmann Meeting, Rio de Janeiro
  - ▷ CMBNet Workshop on Science and Parameter Extraction, Oxford
- 2002 5th Journée de Cosmologie des Lacs Alpins, Annecy
  - ▷ CMBNet Workshop, Geneva
  - ▷ Santa Fe Cosmology Workshop, New Mexico
- 2001 CMBNet General Meeting, Frascati, Italy
  - ▷ UK Cosmology Meeting, Ambleside

## Competitive Grants — Science

Grants above 100k are highlighted in bold. Total value of grants as PI or co-I exceeding 20M €.

1. *Prin ‘Optimal inference from radio images of the epoch of reionization’, Co-I (2022)*. . . . . €192k  
Awarding body: Ministero dell’Università e della Ricerca.
2. *Centro Nazionale Future Artificial Intelligence Research (FAIR), Spoke Edge and Exascale computing, Co-I (2022)*.  
€520k  
Awarding body: PNRR, Italian Government.
3. *Centro Nazionale HPC and Quantum Computing, Spoke COSMOS, Co-I (2022)*. . . . . €565k  
Awarding body: PNRR, Italian Government.
4. *Data Science methods for Multi-Messenger Astrophysics & Multi-Survey Cosmology, PI (2022)*. . . . . €140k  
Awarding body: PRO3 Programma congiunto.
5. *M-Theory, Cosmology and Quantum Field Theory, co-I<sup>†</sup> (2019)*. . . . . £6.4M  
Awarding body: STFC.
6. *Pedagogy Transformation Grant, co-I (2018)*. . . . . £451k  
Awarding body: Education Office, Imperial College London.
7. *M-Theory, Cosmology and Quantum Field Theory, co-I<sup>†</sup> (2016)*. . . . . £6.2M  
Awarding body: STFC.
8. *Astrophysics Consolidated Grant, co-I<sup>†</sup> (2016)*. . . . . £2.4M  
Awarding body: STFC.
9. *Development of Novel Statistical Tools for the Analysis of Astronomical Data, Marie Skłodowska-Curie Research and Innovation Staff Exchange, co-I (2016)*. . . . . £107k  
Awarding body: European Commission.
10. *Impact Acceleration Grant, PI (2016)*. . . . . £46k  
Awarding body: EPSRC.
11. *Impact Acceleration Award, PI (2014)*. . . . . £15k  
Awarding body: STFC.
12. *Pathways to Impact funding, PI (2013)*. . . . . £14k  
Awarding body: EPSRC.
13. *Astrophysics Consolidated Grant, Co-I<sup>†</sup> (2012)*. . . . . **£463k**  
Awarding body: STFC.
14. *Extragalactic Astronomy and Cosmology Rolling Grant, co-I<sup>†</sup> (2009)*. . . . . £1M  
Awarding body: STFC.
15. *Sir Norman Lockyer Fellowship, PI (2005)*. . . . . £150k  
Awarding body: Royal Astronomical Society.
16. *Postdoctoral Fellowship, PI (declined) (2005)*. . . . . £ 90k  
Awarding body: European Space Agency.

## Competitive Grants — Science Communication

17. *Incentivi Triennali alle Iniziative di Divulgazione, PI (2026–2029)*. . . . . €200k  
Awarding body: Regione FVG.
18. *Investigative journalism grant, co-I (2025)*. . . . . €50k  
Awarding body: Journalism Science Alliance.
19. *Bando Divulgazione Cultura Scientifica, PI (2025)*. . . . . €30k  
Awarding body: Regione FVG.
20. *SISSA Summer Festival present: Stelle Scadenti, PI (2021)*. . . . . €25 k  
Awarding body: Regione Autonoma Friuli Venezia Giulia.
21. *The Edge of the Sky storytelling project, co-I (2020)*. . . . . £12k  
Awarding body: Creative Scotland.
22. *Public Engagement Grant, PI (2016)*. . . . . £1.5k  
Awarding body: Royal Astronomical Society.
23. *Public Engagement Grant, PI (2016)*. . . . . £2k  
Awarding body: Institute of Physics.

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<sup>†</sup>For this kind of group-wide grants, the PI is by default the Head of Group, even though all co-I’s contribute equally to the scientific proposal.

24. *Public Engagement Fellowship, PI* (2013). ..... £106k  
Awarding body: STFC.
25. *Public Engagement Grant, PI* (2011). ..... £8.5k  
Awarding body: STFC.

## Travel Grants, Student Fellowships and Conference Support Grants

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26. *Erasmus Mundus programme development support, co-I* (2026). ..... €60k  
Awarding body: EU.
27. *Visiting Scientist Travel Grant, PI* (2026). ..... €4k  
Awarding body: SISSA.
28. *Visiting Scientist Travel Grant, PI* (2024). ..... €10k  
Awarding body: SISSA.
29. *PhD studentship Dottorati Green/Innovazione/DM 351, PI* (2023). ..... €100 k  
Awarding body: PNRR, Italian Government.
30. *PhD studentship ‘PON Innovazione’, PI* (2021). ..... €100 k  
Awarding body: PNRR, Italian Government.
31. *PhD studentship, PI* (2017). ..... £70k  
Awarding body: STFC.
32. *Travel and Exchange Grant, PI* (2016). ..... £12k  
Awarding body: Royal Society.
33. *SPRINT Collaboration Grant, PI* (2016). ..... £10k  
Awarding body: FAPESP-Imperial.
34. *Conference Organization Grant, PI* (2015). ..... £35k  
Awarding body: IFT, Madrid.
35. *Data Analysis Summer School funding, Co-I* (2014). ..... £23k  
Awarding body: STFC.
36. *Search for Dark Matter particles with the XENON experiment collaboration exchange, PI* (2013). ..... £6k  
Awarding body: Imperial College Trust.
37. *PhD studentship, PI* (2013). ..... £70k  
Awarding body: STFC.
38. *Programme Organization Grant, PI* (2013). ..... £65k  
Awarding body: KITP, Santa Barbara.
39. *Travel and Exchange Grant, PI* (2012). ..... £12k  
Awarding body: Royal Society.
40. *CosmoClassic Cross-Disciplinary Collaboration, Co-I* (2012). ..... £16k  
Awarding body: Faculty Strategic Research Fund, Imperial College London.
41. *Conference Organization Grant, Co-I* (2012). ..... £25k  
Awarding body: Centre Stefano Franscini, ETHZ.
42. *Travel Grant, PI* (2011). ..... £1.5k  
Awarding body: Deutsche Studentstiftung.
43. *PhD studentship, PI* (2011). ..... £36k  
Awarding body: Deutsche Studentstiftung.
44. *Summer Student Project Bursaries, PI* (2011). ..... £3k  
Awarding body: UROP.
45. *PhD Studentship Extension Funding, PI* (2011). ..... £3k  
Awarding body: Royal Astronomical Society.
46. *Travel Grant, PI* (2010). ..... £2k  
Awarding body: Royal Astronomical Society.
47. *PhD studentship, PI* (2008). ..... £70k  
Awarding body: STFC.

## Scientific Publications

ORCID ID: [0000-0002-3415-0707](#) | Scopus Author ID: [7004543998](#) | Web of Science ResearcherID: [AAN-7820-2020](#) | iNSPIRE author ID: [R.Trotta.1](#) | [Google Scholar](#)

115 published papers in peer-reviewed physics journals or machine learning conferences (12 of which as first author, including a single-author review article and 3 in *Phys. Rev. Lett.* ); 18 conference proceedings; 5 white papers and editorial works; 5 chapters in academic books.

Citation count on July 28, 2026: 14,000+, h-index: 53 (from [iNSPIRE](#)); 15,000+, h-index 59 (from [Google Scholar](#)).

Co-authors who at the time of publication were my PhD or Master students are underlined.

indicates a paper with more than 100 citations (16).

indicates a paper with more than 250 citations (4).

indicates a paper with more than 500 citations (2).

indicates a paper with more than 1000 citations (3).

100+  
250+  
500+  
1000+

## Refereed Physics Journals

1. J. Aalbers et al (DARWIN Collaboration; A. Scaffidi & **R. Trotta** as corresponding authors, 2026), ‘Model-independent searches of new physics in DARWIN with a semi-supervised deep learning pipeline’, *Eur. Phys. J. C*, **86**, 312 (2026) [Eur. Phys. J. C 86, 312 \(2026\)](#), [arxiv:2410.00755](#).
2. S. Mishra, **R. Trotta** & M. Viel (2026), ‘Cosmo-FOLD: Fast generation and upscaling of field-level cosmological maps with overlap latent diffusion’, submitted, [arxiv:2601.14377](#)
3. M. de los Rios, S. di Gioia, F. Iocco & **R. Trotta** (2025), ‘Dark Matter profiles of "in silico" galaxies: deep learning inference’, submitted, [arxiv:2510.18964](#)
4. K. Karchev, **R. Trotta** & R. Jimenez (2026), ‘CIGaRS I: Combined simulation-based inference from SNaE Ia and host photometry\*’, *Nature Astronomy* in print, [arxiv:2508.15899](#)
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14. **R. Trotta** & G.D. Starkman (2006), ‘What’s the trouble with anthropic reasoning?’, in C. Munoz and G. Yepes, Eds (2006). AIP Conference Proceedings, 878, *2nd International Conference on The Dark Side of the Universe DSU 2006, Madrid (Spain), June 2006*. AIP [arxiv:astro-ph/0610330](https://arxiv.org/abs/astro-ph/0610330)
15. **R. Trotta** (2006), ‘Cosmological Bayesian model selection’, in L. Lyons and M. K. Ünel, Eds (2006). *Statistical Problems in Particle Physics, Astrophysics and Cosmology. Proceedings of PHYSTAT05, Oxford, UK, 12–15 September 2005*, 15–18, Singapore: World Scientific, [doi.org/10.1142/9781860948985\\_0004](https://doi.org/10.1142/9781860948985_0004)
16. **R. Trotta** & R. Durrer (2006), ‘Testing the paradigm of adiabaticity’, in M. Novello, S. Perez Bergliaffa & R. Ruffini Eds (2006). *The Tenth Marcel Grossmann Meeting. On Recent Developments in Theoretical and Experimental General Relativity, Gravitation and Relativistic Field Theories*, 1739-1743, Singapore: World Scientific. [arxiv:astro-ph/0402032](https://arxiv.org/abs/astro-ph/0402032)
17. G. Rocha, **R. Trotta**, C.J.A.P. Martins, A. Melchiorri, P.P. Avelino, R. Bean & P. Viana (2003), ‘New constraints on varying  $\alpha$ ’, *New Astron. Rev.*, 47, 863–869, [arxiv:astro-ph/0309205](https://arxiv.org/abs/astro-ph/0309205)
18. **R. Trotta** (2003), ‘The cosmological constant and the paradigm of adiabaticity’, *New Astron. Rev.*, 47, 769–774, [arxiv:astro-ph/0304525](https://arxiv.org/abs/astro-ph/0304525)

## Book Reviews

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1. **R. Trotta** (2014), ‘Practical Statistics for Astronomers, 2nd Ed (Wall and Jenkins, CUP, 2012)’, *AIAA Journal*, 52, 9, (2014), 2098, [doi.org/10.2514/1.J053573](https://doi.org/10.2514/1.J053573)
2. **R. Trotta** (2012), ‘Modern Statistical Methods for Astronomy (Feigelson and Babu, CUP 2012)’, *Mathematical Reviews*, MR2963292
3. **R. Trotta** (2008), ‘Statistical challenges in modern astronomy IV (ASP Conference Series, Vol 371)’, *The observatory*, May 2008
4. **R. Trotta** (2008), ‘The Oxford book of modern science writing (Richard Dawkins, Ed, OUP 2008)’, *Oxford Today*, 21, 1 (2008), 53

## KNOWLEDGE TRANSFER

### Board Memberships and Advisory Roles

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|         |                                                                                                                                                           |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026–   | Board of Directors, <a href="https://www.fondazioneinternazionale.com/">Fondazione Internazionale Trieste per il Progresso e la Libertà delle Scienze</a> |
| 2025–   | Scientific Advisory Board, <a href="https://www.facta.eu/">FACTA.eu</a>                                                                                   |
| 2024–   | Scientific Advisory Board, RACHAEL S.r.l., <a href="https://www.rachael.swg.it">rachael.swg.it</a>                                                        |
| 2021–24 | Chairman, Scientific Advisory Board, RACHAEL S.r.l., <a href="https://www.rachael.swg.it">rachael.swg.it</a>                                              |

|         |                                                                                                                          |
|---------|--------------------------------------------------------------------------------------------------------------------------|
| 2018–20 | Non-executive Director, Imperial Consultants, <a href="http://imperial-consultants.co.uk">imperial-consultants.co.uk</a> |
| 2012–20 | Director and co-founder, Data Fusion Consultants Ltd (spin-off company)                                                  |
| 2010–   | Consultant, <a href="#">Imperial Consultants</a>                                                                         |

## Areas of Expertise

**Expertise.** Statistical inference and model selection. Data science and machine learning. Numerical data analysis and data mining. Environmental assessment of risk. Astrophysical phenomena, theoretical physics, space travel and cosmology.

**Services.** Data analysis and interpretation. Probabilistic modeling. Inference, prediction and optimization. Training and workshops: probability theory, inference, numerical methods, computational solutions, data science, machine learning. Expert witness. Scientific consulting for museums and creative arts industry.

## Clients

**Scientific Consultancy.** Beluga Group Ltd (2025); Savio Macchine Tessili (2024); Museo del Castello di Miramare, exhibit *Kosmos* (2023); Oxford University Press (2023, credited); Red Planet Pictures, *Death in Paradise* (2022, credited); Altin Homes (2021); Regione Autonoma FVG (2021); Nature Publishing Group (2020); Fantastic Productions, Netflix feature film (2019); Imperial College outreach and Royal Albert Hall (2018); Enchanted Lions Books (2016, 2018, 2019, credited); The Polish Cultural Institute, London (2018); Left Bank Pictures, *Origin* (2018, credited); Institute of Physics (2017); Ore Design Jewellery (2015); BBC *Sky at Night* (2015); UP Projects *The Floating Cinema* (2015); The Times Cheltenham Science Festival (2015); Objective Productions *Breaking Magic 1 & 2* (2014); Cinema.ai (2014); Magnum Media *Man vs Expert* (2014); INFRA (2013); Catsnake Productions *The Theory of Everything* (2012, Audience Award, 29th International Short Film Festival Berlin 2013); London Science Museum (2008); Deutsches Museum Munich (2006).

**Statistical Consultancy.** Clients in the shipping industry, energy industry, property developers, designers, banking and finance, environmental groups, fashion, online commerce.

## Keynotes, Lectures and Training

|         |                                                                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026    | Keynote, Generali Security Council; PKB Academy (Lugano)                                                                                                                |
| 2025    | Keynote, FINECO Advisory Seminar                                                                                                                                        |
| 2024    | Online seminar, Intesa San Paolo                                                                                                                                        |
| 2022    | Lecturer, CINECA Academy, up-skilling programme for businesses, Trieste; Programme committee member and lecturer, Generali Business Translators training course, online |
| 2021    | Knowledge transfer webinar: the case of <i>RACHAEL</i> , a big data startup (online)                                                                                    |
| 2019–21 | Imperial College Business School Executive Education: Sberbank Digital Technology Programme (2020, 2021); Naspers Executives Programme (2019)                           |
| 2018    | Imperial Space Lab STFC Showcase Industry Event                                                                                                                         |
| 2017    | Keynote, Data Analysts User Group conference, London                                                                                                                    |
| 2016    | Keynote, iHub opening ceremony, Imperial College London                                                                                                                 |

## Public Engagement and Science Communication

### Books for the General Public

1. **R. Trotta** (2023), *Starborn. How the Stars Made Us (And Who We Would Be Without Them)*, Basic Books. Translations: Italian (Il Saggiatore, 2025); Korean (Mirae N Co., 2025); Spanish (Pasado & Presente, 2025); Chinese (simplified characters, Unread Sky, 2026). Awards and distinctions: BBC Radio 4 'Book of the Week'; BBC Sky at Night 5-star review; 'The 22 best non-fiction and popular science books of 2023' (New Scientist); 'Ten Best Science Books of 2023' (The Smithsonian Magazine); 'Ten Best Books of December 2023' (Christian Science Monitor); 'Fall STEM Reads of 2023' (American Scientist); 2025 Nautilus Book Awards Gold Winner.
2. **R. Trotta** (2014), *The Edge of the Sky: All You Need to Know about the All-There-Is*, Basic Books. The book explains cosmology using only the most common 1,000 words in English. Translations: Korean (Kyobo Centre Books, 2015), German (C.H. Beck, 2015), Catalan (El Magnanim, 2020). Awards and distinctions: Global Thinker 2014 for 'junking astronomy jargon' (*Foreign Policy*, Nov/Dec 2014); 'Best Science Books of 2014', brainpickings.org; 'Great Popular Science Books 2014', *The Huffington Post*; 'Favorite Physics Books of 2014', *Scientific American*; 'Recommended' by Scientific American (Aug 2014); 'Most anticipated books of 2014', *Publishers Weekly*, July 2014; starred review, *Publishers Weekly*, July 2014.

## Book Chapters and Fiction

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1. **R. Trotta** (2022c), ‘[Wunderlich Park](#)’, *Tamarind Literary Magazine*, Issue 4, Dec 2022.
2. **R. Trotta** (2022b), ‘The Dictionary of Invisible Meanings’, In *Picturing the Invisible*, R. Morgan and P. Coldwell (eds), UCL Press (2022)
3. **R. Trotta** (2022a), ‘The Invisible Universe’, In *Picturing the Invisible*, R. Morgan and P. Coldwell (eds), UCL Press (2022)
4. **R. Trotta** (2021), ‘[Masters and Servants: The Need for Humanities in an AI-dominated Future](#)’, in *The Love Makers*, A. Campbell (ed), Goldsmith Press/MIT Press

## Other publications for the General Public

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1. **R. Trotta** (2025b), ‘Un Mahler di silicio’, Trieste a Teatro, Aprile-Maggio 2025, 40–41.
2. **R. Trotta** (2025a), ‘[The stars have always spoken to us. It’s time to listen again](#)’, Big Issue Magazine, Jan 13th 2025.
3. **R. Trotta** (2024b), ‘[Da Asimov a Musk, il nostro destino è fra le stelle?](#)’, TuttoScienze, La Stampa, Oct 30th 2024.
4. **R. Trotta** (2024a), ‘[In search of lost deep time](#)’, Tangible Territory, Issue 7, Sept 2024.
5. A. Contessa, P. Del Negro, S. Fantoni, R. Rui & **R. Trotta** (2024), ‘Un mare di dati/An ocean of data’, in: *Kosmos — Il veliero della conoscenza/The sailing ship of knowledge*, exhibition catalogue, SilvanaEditoriale (2024), 141–149
6. **R. Trotta** (2023), ‘[What We Lose When We Can’t Stargaze](#)’, *TIME*, Dec 15th 2023
7. **R. Trotta** (2022b), ‘[Verso un futuro dominato dall’intelligenza artificiale](#)’, *Italian Tech*, Nov 2022
8. **R. Trotta** (2022a), ‘Il peso del cielo’, *Meridiana* 276, Mar-Apr 2022, 18–25.
9. **R. Trotta** (2020c), ‘[Vintage 2020](#)’, Tangible Territory, Issue 1, Oct 2020.
10. **R. Trotta** (2020b), ‘[Diary of an astrophysicist](#)’, in *The Covid Diaries: Views from Imperial College*, S. Webster (Ed), online e-zine.
11. **R. Trotta** (2020a), ‘From the Big Bang to AI’, In *Uncertain Ruins*, Petrel, London, (2020)
12. **R. Trotta** (2019b), ‘The Smell of Dark Matter: Dane Mitchell’s *Perfume Plumes*’, *Art in Print*, Vol 9, 3, Sept-Oct (2019)
13. **R. Trotta** (2019a), ‘The Harmony of the Cosmos’, Space Spectacular programme brochure Royal Albert Hall, June 2019
14. **R. Trotta** (2018), ‘The Harmony of the Cosmos’, Space Spectacular programme brochure Royal Albert Hall, May 2018
15. **R. Trotta** (2015f), ‘The Physics of Star Trek’, Star Trek: The Ultimate Voyage programme brochure Royal Albert Hall, Nov 1st 2015
16. **R. Trotta** (2015e), ‘In so many words: Minute world with big stories to tell’, *NewScientist* online, July 30th 2015
17. **R. Trotta** (2015d), ‘In so many words: Don’t kill small-life before Red World trip’, *NewScientist* online, July 24th 2015
18. **R. Trotta** (2015c), ‘Hoped-for dark matter flash might instead be the corpses of stars’, *NewScientist* online, July 1st 2015
19. **R. Trotta** (2015b), ‘In so many words: How to ride the space-wind to the stars’, *NewScientist* online, June 4th 2015
20. **R. Trotta** (2015a), ‘Simple is Hard’, *Army ALT*, Jan-Mar 2015
21. **R. Trotta** (2014d), ‘Is the All-There-Is all there is?’, *Significance* magazine, December 2014
22. **R. Trotta** (2014c), ‘The science of Interstellar: Astrophysics, but not as we know it’, *The Guardian*, Nov 5th 2014
23. **R. Trotta** (2014b), ‘Can the universe be described in simple English?’, British Council’s VOICES invited blog post, Nov 12th 2014
24. **R. Trotta** (2014a), ‘Mangalyaan: The Power of Science and Simplicity’, *The Eastern Eye*, Oct 3rd 2014
25. **R. Trotta** (2013), ‘Only one Universe to observe’, *SCOPE*, May 2013
26. **R. Trotta** (2012), ‘Cosmic Dialogue’, *People and Science*, March 2012 (featured article)
27. **R. Trotta** (2007e), ‘The trouble with strings. Interview with Lee Smolin’, *Newton*, November 2007 (cover story)
28. **R. Trotta** (2007d), ‘Dark Matter: Probing the Arche-Fossil’, *COLLAPSE* Vol II, March 2007
29. **R. Trotta** (2007c), ‘Before the Big Bang’, *Newton*, September 2007 (cover story)
30. **R. Trotta** (2007b), ‘Mapping out the invisible Universe’, *Newton*, April 2007 (cover story)
31. **R. Trotta** (2007a), ‘The anthropic principle: Interview with John Barrow’, *Newton*, October 2006 (cover story)
32. **R. Trotta** (2006g), ‘Cosmologists scoop Nobel Prize for Physics 2006’, *LeScienze*, Nov 2006
33. **R. Trotta** (2006f), ‘Giant telescopes of the future: the Square Kilometer Array’, *LeScienze*, July 2006

34. **R. Trotta** (2006e), ‘Latest news from the baby Universe’, LeScienze, June 2006
35. **R. Trotta** (2006d), ‘Interview with James Lovelock’, LeScienze, April 2006
36. **R. Trotta** (2006c), ‘Listening to cosmic sound’, LeScienze, January 2006.
37. **R. Trotta** (2006b), ‘Modified gravity’, Newton, April 2006
38. **R. Trotta** (2006a), ‘Killer asteroids’, Newton, January 2006
39. **R. Trotta** (2005c), ‘Looking at the Universe through neutrinos’, LeScienze, December 2005
40. **R. Trotta** (2005b), ‘The mysterious past of our Universe’, Newton, October 2005
41. **R. Trotta** (2005a), ‘Einstein’s legacy’, Newton, June 2005

## Public Lectures, Festivals, Educational Activities and Talks in Schools

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|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | Starborn — CERN Library talk <ul style="list-style-type: none"> <li>▷ Keynote, AstroCalina fundraising event</li> <li>▷ Panelist, World Pignolo Day, Udine</li> <li>▷ Keynote, AI-peritivo, Trieste</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2025 | Locarno Palacinema <ul style="list-style-type: none"> <li>▷ Farra d’Isonzo Astronomy Club</li> <li>▷ Genova Science Festival</li> <li>▷ Festival della Scienza dell’Alto Vicentino, Schio</li> <li>▷ PordenoneLegge Book Festival</li> <li>▷ Strange Days, GO25!, Gorizia</li> <li>▷ Lavagne, Festival della Letteratura, Mantova</li> <li>▷ Festival della Letteratura, Mantova</li> <li>▷ Salone del Libro di Torino</li> <li>▷ ‘Cresciuti dalle stelle’, conferenza-spettacolo, Festival del libro scientifico Scienza &amp; Virgola, Trieste</li> <li>▷ ‘Scrivere di scienza’, panelist, Festival del libro scientifico Scienza &amp; Virgola, Trieste</li> <li>▷ ‘Dall’utopia alla tecnocrazia distopica: il ruolo dell’Intelligenza Artificiale nell’ideologia del Muskismo’, AI Talks, Trieste</li> <li>▷ ‘<a href="#">How The Stars Built Our World</a>’, Royal Institution, London</li> <li>▷ Oxford Literary Festival, Oxford</li> <li>▷ Delaware Astronomy Club, online</li> <li>▷ Bookpride, Milano</li> <li>▷ ‘The life of stars’, elementary school Il Castelletto, Trieste</li> <li>▷ ‘<a href="#">Ancient humans and the cosmos</a>’, The Prehistory Guys YouTube channel</li> </ul> |
| 2024 | ‘Da Isaac ad Elon’, MondoFuturo, Trieste <ul style="list-style-type: none"> <li>▷ ‘AI e Astrofisica’, TriesteNext panelist, Trieste</li> <li>▷ MediaLink festival panelist, Trieste</li> <li>▷ Star walk, ALSO 2024 Festival</li> <li>▷ ‘Starborn’, ALSO 2024 Festival</li> <li>▷ ‘AI for Science’ public debate moderator, University of Amsterdam</li> <li>▷ Festival della Scienza dei Dati e dell’Intelligenza Artificiale (Monfalcone)</li> <li>▷ Festival della Scienza e Filosofia (Foligno)</li> <li>▷ Scuola Media Galilei, Sant’Eraclio</li> <li>▷ Grand Incontri, Miramare Castle (Trieste)</li> <li>▷ AI from the Sky, United World College (Duino)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2023 | <a href="#">Q&amp;A with Louise Allcock and Robin Ince</a> , RATIO Forum (Sofia) <ul style="list-style-type: none"> <li>▷ Starborn — <a href="#">How the Stars Made Us</a>, RATIO Forum (Sofia)</li> <li>▷ Starborn — book launch, in conversation with Claudia de Rham, Imperial College London (London)</li> <li>▷ Starborn — book launch, in conversation with Roger Davies, St Anne’s College (Oxford)</li> <li>▷ Starborn — book launch, in conversation with Carissa Vêliz, Hertford College (Oxford)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

- ▷ From Babbage to Musk: Promethean Visions of AI, keynote, MIB Trieste School of Management alumni reunion (Trieste)
  - ▷ [Dal Big Bang all'Intelligenza Artificiale](#), QuarantaScienza, Accademia dei XL (online)
  - ▷ [ChatGPT, do you love me?](#) Panel chair at TriesteNext
  - ▷ I promemoria di Italo Calvino per affrontare il terzo millennio, discussion panel, Scienza e Virgola festival (Trieste)
  - ▷ Dal Big Bang all'Intelligenza Artificiale, AITalks series (Trieste)
  - ▷ [Dal Big Bang all'intelligenza artificiale](#), I venerdì dell'universo (Ferrara)
  - ▷ Il sistema solare, Scuola Europea "Il Castelletto" (Trieste)
- 2022 [Communicating AI, and communicating with AI: challenges and opportunities](#), Future visions on science communication (Trieste)
  - ▷ La tua vita in dati, i dati della tua vita, SISSA4Schools (Trieste)
  - ▷ [Dal Big Bang all'Intelligenza Artificiale](#), Biennale di Tecnologia (Torino)
  - ▷ Scelte intelligenti? Come l'intelligenza artificiale entra nella nostra vita, CICAP Fest (Padova)
- 2021 Il cielo di domani, fra satelliti e stelle, workshop al Convegno Nazionale di Comunicazione della Scienza (Trieste)
  - ▷ G-astronomy: A culinary and multi-sensorial exploration of the Universe (online)
  - ▷ "Stelle Scadenti", Scienza e Virgola science festival, Trieste
  - ▷ Understanding the Universe with AI, Loughton Astronomical Society (online)
  - ▷ Cosmo e dati, panel discussion for SISSA Schools open day (online)
  - ▷ Understanding the Universe with AI, St John's College Oxford (online)
- 2020 [From the Big Bang to AI](#), Imperial Science Breaks, online
  - ▷ Big Bang Data, ESOF 2020, Trieste
  - ▷ SISSA Summer Science Festival, Miramare Castle, Trieste
  - ▷ [Language of the stars](#), super/collider (online)
  - ▷ The Solar System for Kids, Building Blocks Nursery, Wimbledon, London
  - ▷ G-Astronomy Cosmic Dinner, Uig Sands restaurants, Uig bay
  - ▷ [G-Astronomy Cosmic Cocktails and Canapés](#), An Lanntair restaurant, Stornoway
  - ▷ [From the Big Bang to AI](#), Professorial Inaugural Lecture, Imperial College London
- 2019 Keynote, Web Summit DeepTech Stage, Lisbon
  - ▷ Liceo Cantonale di Locarno, Talk for the general public, Switzerland
  - ▷ Liceo Cantonale di Locarno, Talk to high school students, Switzerland
  - ▷ Herstmonceux Science Festival
  - ▷ Spotlight talk, Great Exhibition Road Festival, London
  - ▷ [Multi-Sensorial Dark Matter Experience](#), Great Exhibition Road Festival, London
  - ▷ Poplar Primary School Space Day (main organizer), Merton Park, London
  - ▷ Multi-sensorial Dark Matter Experience, Science Museum Lates, London
- 2018 New Scientist Live, London
  - ▷ [Dining with Copernicus](#), immersive dining experience, Ognisko Restaurant/Polish Cultural Institute, London
  - ▷ STEM workshop for high ability students with high-functioning autism/Asperger's, Imperial College London
  - ▷ Public Engagement Day Pecha Kucha, Imperial College London
  - ▷ The Science of Star Trek, Royal Albert Hall
  - ▷ Stephen Hawking's Universe explained in 1,000 simple words, El Debats del Magnànim, Valencia
  - ▷ Craters on the Moon, Early Years Education Centre, Imperial College London
  - ▷ The Alternative Guide to STEM, Queen's Park Community School, London, winner of audience vote
  - ▷ The Map of the Universe, Data Science Institute, Imperial Festival
  - ▷ [The Science of Star Trek](#), Imperial Festival
- 2017 EdFoo Camp, Google HQ, Mountain View
  - ▷ [Il Tempo nell'Universo](#), Università della Svizzera Italiana, Lugano, Switzerland
  - ▷ [g-ASTRONOMY for people with visual impairment](#), London
  - ▷ JV Parekh International School, Mumbai, India

- ▷ Science City Auditorium, Ahmedabad, India
  - ▷ IUCAA, Pune, India
  - ▷ Birla Planetarium, Chennai, India
  - ▷ Second Home cultural programme, London
  - ▷ CLCC Research Seminar, Imperial College London
  - ▷ Being Human Festival, London
- 2016 g-ASTRONOMY, Cheltenham Science Festival
  - ▷ Panel: To Infinity and Beyond, Imperial Festival, Imperial College London
  - ▷ g-ASTRONOMY, Imperial Festival, Imperial College London
  - ▷ Early Years Education Centre, Imperial College London
  - ▷ Jersey School for Girls, Jersey
  - ▷ St Mary's Primary School, Jersey
  - ▷ Jersey Astronomy Club, Jersey
  - ▷ Why Society Needs Astronomy and Cosmology, Guest Gresham College Lecture, Museum of London
  - ▷ A Matter of Gravity, Royal Astronomical Society public lecture
  - ▷ Big Bang! Daesbury Laboratory
- 2015 g-ASTRONOMY: the cosmos at the tip of your tongue, Science Museum Lates, London
  - ▷ Invited talk, Guadalajara International Book Fair, Mexico
  - ▷ Panel: Science in the UK, Guadalajara International Book Fair, Mexico
  - ▷ Panel: The Language of Science, Guadalajara International Book Fair, Mexico
  - ▷ Exploring the Universe, Guadalajara International Book Fair, Mexico
  - ▷ Panel: What scientists can gain from literature, Guadalajara International Book Fair, Mexico
- Mexico
  - ▷ International Astronomy Colloquium Round table, Guadalajara International Book Fair, Mexico
- ▷ Creative Quarter invited talk, London
  - ▷ The Physics of Star Trek, Royal Albert Hall, London
  - ▷ The Blackett Colloquium, Imperial College London
  - ▷ Imperial Horizons guest lecture, Imperial College London
  - ▷ Across RCA workshop, Imperial College London
  - ▷ Interactive cosmology workshop, Technopop Brixton
  - ▷ Invited talk, Science For Fiction 2015, Imperial College London
  - ▷ Imperial College London Open Day talk
  - ▷ Invited lecture, Graveney School Physics Society
  - ▷ Lightning talk, Google SciFoo Camp, Mountain View
  - ▷ Keynote speech, Graduate School Symposium, Imperial College London
  - ▷ Imperial Open Day talk, Imperial College London
  - ▷ A-level school visit, Imperial College London
  - ▷ Book tour talk, Zurich Dalkey Book Festival, Dublin
  - ▷ The Great Cosmic Cookery Show, The Time Cheltenham Science Festival
  - ▷ Performance at art show opening, Institute for Contemporary Art, London
  - ▷ Book tour event, Imperial Festival
  - ▷ Big Bang Bash event, Edinburgh International Science Festival
  - ▷ Book tour talk, Edinburgh International Science Festival
  - ▷ Soapbox talk, Edinburgh International Science Festival
  - ▷ Invited lecture, Airware, San Francisco
  - ▷ Invited talk, Royal Institution
  - ▷ Book tour talk, Imperial Fringe Festival
  - ▷ Invited talks, Winchester planetarium
- 2014 Book tour talk, Commonwealth Club of California, San Francisco
  - ▷ The Edge of the Sky, Authors at Google, Mountain View
  - ▷ Book tour talk, Town Hall Seattle, Seattle
  - ▷ Book tour talk, Powell's Bookstore, Portland, OR
  - ▷ Public lecture, Technopop, Stratford London

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|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <ul style="list-style-type: none"> <li>▷ Invited lecture, Manchester Science Festival, Manchester</li> <li>▷ Invited lecture, World Teach In weekend, London</li> <li>▷ Invited lecture, English Language Council lecture, London</li> <li>▷ Multiple public lectures, Imperial Fringe Festival, London</li> <li>▷ Exhibitor, World Science Fiction Convention 2014, London</li> <li>▷ IOP public lecture, Salisbury</li> <li>▷ Public lecture, Imperial College London</li> <li>▷ Great Cosmic Cookery Show, Imperial Festival</li> <li>▷ Invited talk, The Observatory, Hertsmonceaux astronomy festival</li> <li>▷ Invited talk, Science for Fiction workshop, London</li> <li>▷ Science workshop for families, Cosmic Fun Club, Glasgow Science Centre</li> <li>▷ Public talk, The Great Cosmic Cookery Show, Imperial Festival, London</li> <li>▷ <a href="#">Science workshop for young children, MoonBerry Muffins!</a>, Winchmore Hill</li> <li>▷ <a href="#">Our place in the multiverse</a>, TEDxHackney ‘Radical Beauty’</li> <li>▷ Invited lecture, Alleyn’s School, London</li> <li>▷ Invited lecture, St Paul’s Way Trust School, Tower Hamlet</li> <li>▷ Multiple invited classroom activities, Norland Place School, London</li> <li>▷ Classroom activity, St. Joseph’s Primary School, London</li> <li>▷ Invited talk, Bishop Challoner Catholic Collegiate School, London</li> </ul> |
| 2013 | Invited lecture, The London Oratory School, London <ul style="list-style-type: none"> <li>▷ Invited talk, Wimbledon High School, Wimbledon</li> <li>▷ Invited talks, MENSA Society, Birmingham</li> <li>▷ Public talk, SPACE at the White Building, London</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2012 | Invited talk, AstroSoc, Imperial College London <ul style="list-style-type: none"> <li>▷ Public talk, Università della Svizzera Italiana, Lugano, Switzerland</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2011 | Invited talk, Friends of the RAS, London <ul style="list-style-type: none"> <li>▷ Public talk, Crayford Manor astronomical society</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2010 | Invited talk, Westminster School, London                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2009 | Invited talk, Marlborough College <ul style="list-style-type: none"> <li>▷ Public talk, HGS Astronomical Society</li> <li>▷ Public talk, Newbury Astronomical Society, Newbury</li> <li>▷ Multiple invited lectures, Festival de Fleurance, France</li> <li>▷ Public talk, Centro Stefano Franscini, Ascona, Switzerland</li> <li>▷ Public talk, INTECH Planetarium, Winchester</li> <li>▷ Public talk, Daresbury Laboratory, Warrington</li> <li>▷ Multiple invited lectures, Kassel High School, Kassel, Germany</li> <li>▷ Exhibitor, Royal Society Summer Science Exhibit</li> <li>▷ Invited lecture, AA School of Architecture, London</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2008 | Invited talk, Association for Astronomy Education’s Annual General Meeting, London <ul style="list-style-type: none"> <li>▷ Public talk and panel discussion, EuroScience Open Forum, Barcelona</li> <li>▷ Family event, Meet the Scientist! Museum of Science and Industry, Manchester</li> <li>▷ Invited talk, Tudor Hall School, Banbury</li> <li>▷ Public lecture, Winchester Planetarium</li> <li>▷ Public talk, CornerClub, Oxford</li> <li>▷ Invited talk, Green College, Oxford</li> <li>▷ Invited lecture, National Particle Physics Masterclass at Oxford University</li> <li>▷ Multiple lectures, Queen Mary 2 Caribbean Adventure voyage</li> <li>▷ Public lecture, Oxford University Astrophysics Department</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2007 | Invited lecture, Oxford University Space and Astronomy Society, Oxford <ul style="list-style-type: none"> <li>▷ Invited talk, Bedford school, Bedford</li> <li>▷ The BA Award Lectures: Lord Kelvin Lecture 2007, York</li> <li>▷ Public lecture, Royal Astronomical Society, London</li> <li>▷ Invited lecture, Marlborough College Summer School</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2006 | Lead exhibitor, Royal Society Summer Science Exhibit, London                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <ul style="list-style-type: none"> <li>▷ Invited activity, Gonville Primary School</li> <li>▷ Invited lecture, Mansfield College, Oxford</li> <li>▷ Invited activity, Sonning Common primary school</li> </ul> |
| 2005 | Invited talk, Bedford school <ul style="list-style-type: none"> <li>▷ Public lecture, CERN, Geneva</li> <li>▷ Invited lecture, Astronomy club of Divonne, France</li> </ul>                                    |
| 2004 | Public talk, CERN, Geneva                                                                                                                                                                                      |

## Science Communication Teaching and Training

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| 2025 | Invited panelist, ‘Giornalismo ed intelligenza artificiale’, training day for professional journalists, Trieste                                                                                                                                                                                                                                                           |
| 2024 | Lecturer, ‘Intelligenza Artificiale: Guida al Cambiamento’, training programme for school teachers, Gradisca                                                                                                                                                                                                                                                              |
| 2020 | Lecturer, ‘Comunicare la ricerca’, Science Communication Masters programme, SISSA                                                                                                                                                                                                                                                                                         |
| 2018 | Workshop leader, ‘The Sound of Space’, Imperial College London and Royal Albert Hall <ul style="list-style-type: none"> <li>▷ Public Engagement Academy Panel member, Imperial College London</li> <li>▷ Public Engagement for Research Fellows, Imperial College London</li> <li>▷ Communicating Science guest lecture MSc programme, Imperial College London</li> </ul> |
| 2017 | Science Communication Lecture, ISAPP Summer School <ul style="list-style-type: none"> <li>▷ Communicating Science guest lecture, Imperial College London</li> </ul>                                                                                                                                                                                                       |

## Courses for the General Public

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| 2021–22 | Gresham College Visiting Professor of Cosmology: “The frontiers of knowledge” (online and in person): <a href="#">The Future of Life on Earth</a> <ul style="list-style-type: none"> <li>▷ <a href="#">The Broken Cosmic Distance Ladder</a></li> <li>▷ <a href="#">Einstein’s blunder</a></li> </ul>                    |
| 2020–21 | Gresham College Visiting Professor of Cosmology: “The unexpected Universe” (online): <a href="#">Space sounds: the music of the cosmos</a> <ul style="list-style-type: none"> <li>▷ <a href="#">Neutrino: the particle that shouldn’t exist</a></li> <li>▷ <a href="#">Understanding the Universe with AI</a></li> </ul> |
| 2019–20 | Gresham College Visiting Professor of Cosmology: “The nature of reality” (Museum of London): <a href="#">Mysteries of the dark cosmos</a> <ul style="list-style-type: none"> <li>▷ <a href="#">What has Einstein ever done for you?</a></li> <li>▷ <a href="#">Weighing the Universe</a></li> </ul>                      |
| 2014    | Invited lecture, 36th Annual Astronomy Weekend, Oxford                                                                                                                                                                                                                                                                   |
| 2010    | Day course (with Matthew Malek), Oxford University Continuing Education                                                                                                                                                                                                                                                  |
| 2009    | Day course (with Bob Lambourne), Oxford University Continuing Education <ul style="list-style-type: none"> <li>▷ Week-long cosmology course, Oxford University Summer School for Adults</li> <li>▷ Multiple lectures, Oxford University Summer School</li> </ul>                                                         |
| 2008    | Weekend course, multiple lectures, 30th Annual Astronomy Weekend <ul style="list-style-type: none"> <li>▷ 20-lectures series on cosmology, Oxford University Continuing Education department</li> <li>▷ Day course (with Bob Lambourne), Oxford University Continuing Education department</li> </ul>                    |
| 2007    | Day course (with Bernard Carr), Oxford University Continuing Education department                                                                                                                                                                                                                                        |
| 2006    | 20-lectures series on cosmology, Oxford University Continuing Education department                                                                                                                                                                                                                                       |

## Art / Science and Data Journalism Projects

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|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | Scientific advisor, <i>La storia dell’Universo</i> , permanent exhibit, SISSA                                                                                                                                                                                                                        |
| 2026 | Scientific advisor, <i>The unwetlands</i> , A Satellite-Based Investigation into Italy’s Lost Wetlands, with FACTA.eu (Elisabetta Tola and others)                                                                                                                                                   |
| 2023 | Producer and scientific advisor, <i>Parl-IA-moci</i> , an original theatre play by and with Diana Höbel, featuring an impossible dialogue between Alma Mahler (played by Diana) and Gustav Mahler (played by ChatGPT). AI consultant. Premiered at the Modena Festival di Filosofia 2023 (Sept 2023) |

- ▷ Soroptimist Club, Trieste (May 2024)
  - ▷ Scienza & Virgola Festival, Trieste (May 2024)
  - ▷ Invisible Cities, Gorizia (Aug 2024)
  - ▷ Teatro Rossetti, Trieste (Apr)
- 2023 Scientific advisor, *Five centuries in five minutes*, in collaboration with Gigi Funcis, an AI-generated video for the exhibit *Kosmos*, Miramare Castle, Trieste
- 2022 Co-author and creative advisor., *The Edge of the Sky / Oir Nan Speur: A theatre show in Gaelic and English*, by production company sruth-mara. Premiere in 2022 at the Dark Sky Festival, Stornoway, Scotland
- 2021–22 Featured in *Ophelia in Exile* by Dr Tereza Stehlikova, multimedia art installation, Czech Centre London
- 2021 Artistic director, concept and script, *LIBRA*, an original multimedia theatre play (with director and video-artist Gigi Funcis), commissioned by SISSA Summer Festival (2021). Nationwide media coverage, including *Il Messaggero*, *Il Corriere della Sera*, *Radio Rai 1 FVG*, *Sky TG24*, *Il Sole 24 ore*, *Il Piccolo*, *RAI Radio 3*, *Rete4 local TV*, etc. Performed at Miramare Castle, Trieste (Sept 2021; Sept 2022), Gradisca (Oct), Reggio Calabria (opening of Cosmos Festival, Oct 2022), Gorizia (part of GO25!, European Capital of Culture, 2025)
  - ▷ Featured in *SISSA hosting*, an immersive audio installation part of the *Invisible Cities Festival 2021*
- 2019 Lead designer, [A Multi-Sensorial Dark Matter Experience](#): in collaboration with human-machine interaction experts, a voyage for two using all five senses to ‘perceive’ dark matter, Great Exhibition Road Festival, London
- 2018 *Dinner with Copernicus*, immersive dining experience (with Lab Collective, The Polish Cultural Institute and Ognisko Restaurant), London
- 2017 Lead designer, *g-ASTRONOMY*: in collaboration with molecular gastronomy chef Josez Youssef and the Royal National Institute for the Blind, a tactile, olfactory and edible experience illustrating cosmological, London
- 2015 Co-author, *All There Was* (with artists O. Hagen and D. Cheeseman), art show at the Institute for Contemporary Art, part of *fig-2*
- 2014 Featured in *The All-There-Is*, short documentary by filmmaker L. Melesio Friedman, winner of “Best Portrait of the Life of a Physicist” award of the FQXi Video Competition (2014), [tinyurl.com/o6edrms](https://tinyurl.com/o6edrms)
- 2013 Scientific advisor, *The Theory of Everything* short film, winner of the audience award, Viral Video Awards at the 29th International Short Film Festival Berlin (2013)
  - ▷ Co-proposer, *The Post-Newtonian Orrery* (with artists O. Hagen and D. Cheesman), 100 best Artangel proposals in
- 2011–2012 Scientific advisor, *Urban Sputnik* (by designer V. Harden), exhibited at the Royal Institution, Royal Astronomical Society, Imperial College and Kinetica art fair (2011–2012)
- 2009–2011 Co-author, *Beyond Entropy* research programme (with architect J. Löffler and artist P. Liversidge), exhibited at the Venice Architecture Biennale, the Milano Triennale and the London Architectural Association (2009–2011)

## Other Activities

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Jury member, *Premio Cosmos*, Reggio Calabria (2022–); International FameLab semi-final judge, Cheltenham (2015); Project advisory board, ‘Dark Matters: an interrogation of thresholds of (im)perceptibility through theoretical cosmology, fine art and anthropology of science’, U. Lancaster (2014–2016); Coordinator, Astrophysics Group outreach activities, Imperial College London (2008–2015).

## Broadcast Media Appearances

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- 2026 ‘Terra chiama terra’ RAI FVG, Apr 28th
  - ▷ Countercurrent podcast, with Roger Kneebone, Jan 12th
- 2025 ‘Mille voci’ solo interview, RSI Rete 1, Dec 10th 2025
  - ▷ ‘What a wonderful world’ RAI FVG Radio, Nov 20th
  - ▷ Retesette TV, May 15th
  - ▷ *Il Giardino di Albert*, Swiss National TV, Mar 14th, 2025
  - ▷ *Radio3 Scienza*, RAI National Radio, March 10th
- 2024 *The Infinite Monkey Cage — Starless Worlds*, BBC Radio 4, Nov
  - ▷ *How to Academy — How the stars made us*, with Robin Ince, Nov
  - ▷ RAI RadioScienza, Oct 29th

- ▷ [Calanais Conversations](#), Oct
  - ▷ [Aula di Scienze: Turing e l'AI nella ricerca scientifica](#), May 30th
  - ▷ [Solar Eclipse Magic](#), in: *Living on Earth*, March 15th 2024
  - ▷ Buongiorno Regione, RAI FVG, Feb 20th
- 2023 [Start of the Week](#), BBC Radio 4, Nov
  - ▷ [Economy.bg](#), Nov 2023
  - ▷ KPCW-Cool Science Radio, Nov 2023
  - ▷ Nova TV (Bulgaria), Nov 2023
  - ▷ Bulgaria National TV, Nov
  - ▷ ReggioTV, Telegiornale, Oct 7th
  - ▷ [Millevoci](#), Swiss National Radio, Rete 1 Sept 26th 2023
  - ▷ TGR Friuli Venezia-Giulia, Sept 23rd
  - ▷ Sei di sera, RSI Rete 1, Aug 23rd
  - ▷ Radio 3 Scienza, Italian National radio, May
  - ▷ [In conversation with with Melanie Mitchell](#), Premio Cosmos Studenti, Apr 2023 (online)
- 2022 [Telegiornale delle 12.30](#), RSI Swiss National TV, Oct 12th
  - ▷ Sei di sera, RSI Rete 1, July 12th
  - ▷ In conversation with Jimena Canales, Premio Cosmos Studenti, Apr 2022 (online)
  - ▷ [Hebridean Dark Skies Festival podcast](#), Jan 21st
- 2021 [Sky TG24](#), Sept 9th 2021
  - ▷ Radio Capodistria, Sept 9th 2021
  - ▷ [Radio3 Scienza](#), Italian national radio, Sept 6th 2021
  - ▷ Live interview, RTL 102.5 radio, Sept 5th 2021
  - ▷ Live appearance, *Tele4* local TV, Aug 31st 2021
  - ▷ [RAI FVG Giornale Radio](#), May 28th 2021
- 2020 [Beyond the Fourth Floor](#), Imperial College students podcast, online
  - ▷ [Retequattro Estate](#), Local TV interview, Trieste, Italy
- 2019 [Millevoci: Solo interview](#), Swiss National Radio, Rete 1
  - ▷ [Millevoci: Buchi Neri](#), Swiss National Radio, Rete 1
- 2018 [Millevoci: Stephen Hawking](#), Swiss National Radio, Rete 1
  - ▷ [Telegiornale Edizione delle 20](#), RSI Swiss National TV
  - ▷ [Millevoci: Hubble e le prime stelle](#), Swiss National Radio, Rete 1
  - ▷ [Millevoci: Da dove nasce il tempo?](#), Swiss National Radio, Rete 1
- 2017 [Millevoci: Onde Gravitazionali](#), Swiss National Radio
  - ▷ [Telegiornale Edizione delle 20](#), RSI Swiss National TV
  - ▷ [Science Mixtape](#), Soho Radio
  - ▷ [Chiacchiere Cosmologiche](#), Swiss National Radio, Rete 3
  - ▷ [Millevoci](#), Swiss National Radio, Rete 1
- 2016 [Mission Impossible](#), ICradio show
  - ▷ [Albachiara](#), Radio Svizzera Rete 1
  - ▷ [Millevoci](#), Swiss National Radio, Rete 1
- 2015 [Einstein's Extraordinary Universe](#), ABCcatalyst Australia
  - ▷ [Neil Delamere's Sunday Best](#), Today FM, Dublin
  - ▷ [The Sky at Night](#), BBC 4
  - ▷ [Sierra Club Radio](#) with Orli Cotel
- 2014 [Albachiara](#), Radio Svizzera Rete 1
  - ▷ [Al Jazeera Arabic](#), news report, Nov
  - ▷ [Inquiry](#), WICN public radio, with Mark Lynch
  - ▷ [Between the Covers](#), KBOO FM, with Leigh Anne Kranz
  - ▷ [Living on Earth](#), with Steve Curwood
  - ▷ [Words on a Wire](#), KETP radio, with Daniel Chacòn
  - ▷ [The Jason Mohammad Show](#), BBC Radio Wales
  - ▷ [Talk Radio Europe](#), with Hannah Murray
  - ▷ [C-SPAN2 Book TV](#), talk at the Commonwealth Club of California

- ▷ *KATU-TV am northwest*, with Helen Raptis and Dave Anderson
  - ▷ *NPR's Weekend Edition Sunday*, with Wade Goodwyn
- 2010 *Baobab*, Swiss National Radio Rete 3
- 2009 *Il Giardino di Albert*, Swiss National TV RSI
  - ▷ *Mille voci*, Swiss National Radio, Rete 1
- 2008 *Future Generation Thinkers*, BBC Radio 3

## Coverage in Print and Online

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- 2025 *War Room InnovAction*, with Luca de Biase, Apr 2025
  - ▷ *Restorations Conversations* magazine, Issue 7, Spring 2025
  - ▷ *Scienzainrete* interview, Mar 21st 2025
  - ▷ *RSI Info* website, Mar 16th 2025
  - ▷ *Il Piccolo* newspaper, Jan 10th
- 2024 *Meridiana* 291, Sett-Ott 2024, 12–15
  - ▷ *Let's Talk: Learning About AI Through Art*, Aspen Institute blog, Oct 25h
  - ▷ *E se non ci fossero le stelle?*, Mangrovia magazine, Sept 9th
- 2022 *The Naked Scientist: Can AI help us understand dark energy better?*, podcast, Sept 30th
  - ▷ *Le Scienze*, March 16th
- 2021 *Il Piccolo* newspaper, Apr 28th
  - ▷ *Dalla Cosmologia al Razzismo*, con Ruggero Rollini, online video
- 2020 *Il Piccolo* newspaper, Oct 27th
- 2018 *CrowdScience*, *Why Does Dark Matter, Matter?*, BBC World Service, podcast
- 2017 *Countercurrent* podcast, with Roger Kneebone
  - ▷ *Scientists, not the Science* podcast, with Stuart Higgins
- 2016 *Institute of Physics* podcast, with James Dacey
  - ▷ *Jersey Evening Post* feature article, Apr 23rd
  - ▷ *reporter* Imperial College London magazine interview
  - ▷ *A New Language For Science*, Science Writing Resources for Learning blog
- 2015 *The Reading Survey*, Edinburgh Science Festival
  - ▷ *Fascination* magazine interview, Summer 2015
  - ▷ fig. 2 “All There Was” podcast, with Jessica Temple
  - ▷ *FQXi* podcast, with Sophie Hebden
  - ▷ *Fiat Physica* online interview
  - ▷ *Book Q&A*, with Deborah Kalb
- 2014 *The Guardian Astronomy Science Weekly* podcast, with Nicola Davies
  - ▷ *Institute of Astrophysics of the Canary Islands* podcast
  - ▷ *Authors at Google*, video of talk at Google HQ
  - ▷ *Il Caffè*, Swiss Italian newspaper interview
  - ▷ *popularscience.co.uk* blog post
  - ▷ *Science for the People* podcast, with Desiree Schell
  - ▷ *Speaking of Science* podcast, with Julie Gould
  - ▷ *Can the universe be described in simple English?* British Council's VOICES blog
  - ▷ *Living on Earth* podcast, with Steve Curwood
  - ▷ *New Books in Physics* podcast, with Meg Rosenberg
  - ▷ *In Residence* podcast, with Steve Scher
  - ▷ *Imperial Podcast*, with Gail Wilson
  - ▷ *Nature* podcast, with Elizabeth Gibney
- 2013 *SCOPE* magazine interview
- 2012 *Pod Academy* podcast
  - ▷ *IOP 100-seconds physics* videos